

claude-windows / 144a31b7-7962-4fee-9d6f-8a7854daa63f

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\144a31b7-7962-4fee-9d6f-8a7854daa63f\subagents\workflows\wf_134043cd-8c3\agent-a7bf2763124487711.jsonl`
- SHA-256: `ffc8155eeb6441ac618b271c5dbde4dbf51a6467ad022af79caf96a20ed1e4bd`
- Source modified: `2026-06-20T09:01:52+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_134043cd-8c3`
- session_id: `144a31b7-7962-4fee-9d6f-8a7854daa63f`
```

Transcript

1. user (2026-06-20T08:59:23.222Z)

IMPORTANT grounding (verified this session): the engine is now DECOUPLED ? pluggable StorageBackend (local + S3/GCS/DO Spaces) in backend/storage/, and a worker `python -m backend.worker {infer|train}` (backend/worker/__main__.py) that pulls from storage -> runs the pipeline -> pushes results. Validated e2e on the CPU droplet 168.144.126.176 reading/writing DO Spaces with the GPU MOCKED (stub detector, RALLY_DETECTOR_IMPL=stub). BOTH the khelsutra site (/srv/khelsutra-site) AND the engine (~/rally) already live on that ONE droplet. The droplet is CPU-only (no GPU) and a SHARED testbed (MinIO 9000/9001, GCS emu 9023) ? do not propose touching those. Real GPU WASB (NativeWasbRunner, "M3.5") is the NEXT build, not done. The FastAPI has NO auth (CORS '*'). EXCLUDE any path under .claude/worktrees/ ? those are stale copies; read canonical docs/ + backend/ only.

Map hosting / exposure / locality / desktop (cwd = C:\\Users\\avidu\\Projects\\badminton-highlight-indexer). Read docs/HOSTING_PLAN.md, docs/PLATFORM_ARCHITECTURE.md, docs/VIDEO_LOCALITY_MODEL.md, docs/DESKTOP_APP_PLAN.md, docs/SETUP_NEW_MACHINE.md, and deploy/digitalocean/*.sh. Report the intended auth/exposure model, the video-locality/privacy model, the ACTUAL droplet/machine layout (paths, ports, GPU vs CPU, secrets), whether a "locally installed" app is already envisioned (and its stack), and the single-tenant guardrails that constrain a publicly-reachable single box. Quote path:line. Return the schema.

2. assistant (2026-06-20T08:59:26.320Z)

I'll map the hosting, exposure, locality, desktop, and droplet layout for this project. Let me start by pulling the repo fresh and reading the relevant docs.

3. user (2026-06-20T08:59:31.145Z)

```
fatal: Cannot fast-forward to multiple branches.
---BRANCH---
feat/native-wasb-runner
```

4. user (2026-06-20T08:59:31.409Z)

```
docs/DESKTOP_APP_PLAN.md
docs/HOSTING_PLAN.md
docs/PLATFORM_ARCHITECTURE.md
docs/SETUP_NEW_MACHINE.md
docs/VIDEO_LOCALITY_MODEL.md
---DEPLOY---
README.md
bootstrap.sh
gpu_setup.sh
mount_scratch.sh
```

5. assistant (2026-06-20T08:59:33.786Z)

Branch can't fast-forward (diverged) ? I'll proceed read-only on the local checkout as instructed. Let me read all the target docs and deploy scripts.

6. user (2026-06-20T08:59:34.743Z)

```
1      # Hosting Plan ? khelsutra.guru
2
3      **Status:** PROPOSAL ? awaiting owner ratification.
4      **Date:** 2026-06-10 . **Constraints:** budget $10?25/mo . GCP credits available .
DigitalOcean
5      acceptable . alpha backend = owner's GPU box behind a tunnel . 10?25 alpha testers .
registrar = GoDaddy.
6      **Companion:** `docs/UI_PROPOSAL.md`.
7
8      ## 1. Recommendation in one line
9
10     **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11     **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**
(Cloud Run +
12     GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
13
14     Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15     Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16     exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17     a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18     Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20     ## 2. Alpha topology ($0/mo)
21
22     ```
23     tester browser
24     ? https
25     ?
26     Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27     ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28     ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30     ?
31     ?
32     FastAPI :8000 on the GPU box (Windows)
33     ? Gemini segmenter path (no GPU in request path)
34     ? async jobs (#16), SQLite WAL, output/ on disk
35
36     - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP
never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38     `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40     chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42     offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44     domain renewal.
```

```

45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ```
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ```
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,
no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),
Cloud Run
61     handles Gemini-path jobs entirely in-cloud.
62     - Auth graduates to **Firebase Auth** (free; Google + email sign-in; India-friendly
phone OTP
63     later); Access allowlist retires.
64     - Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-
to-zero) .
65     GCS 100 GB ? $2.60 + egress $0.12/GB (reel downloads ? watch this line) . Cloud SQL
smallest
66     ? $10 ? **~$15?20/mo**, inside budget. Egress lever if it grows: serve reels from
Cloudflare R2
67     (zero egress) while keeping everything else GCP.
68
69     ## 4. Alternatives considered
70
71     | Option | Monthly (alpha ? beta) | Pros | Cons |
72     |---|---|---|---|
73     | **Recommended hybrid** (CF edge + box ? +GCP) | **$0 ? ~$0 w/ credits** (~$15?20
after) | Cheapest at every phase; credits used where they matter; maps to PLATFORM P2/P3 | Two
vendors (CF + GCP) |
74     | **All-DigitalOcean** | ~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB + **1 TiB egress
included**) | One simple vendor; Spaces' included egress is the best video-download deal; per-
second billing | No credits; no scale-to-zero; still needs a tunnel-equivalent to reach the box
(run `frp`/WireGuard on the droplet) |
75     | **All-GCP from day 1** | ~$18+ ? credits | Single vendor, credits immediately | LB+IAP
standing cost just to gate an alpha; no free tunnel to the home box; credits burn on plumbing |
76     | **All-Cloudflare** (Pages+Workers+R2) | ~$0 ? ~$5 | Zero egress on R2; cheapest
forever | FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP
credits |
77
78     ## 5. Domain setup ? GoDaddy ? Cloudflare DNS (one-time, ~15 min, you)
79
80     1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan.
Cloudflare
81     imports existing records and shows **two nameservers** (e.g.
`xxx.ns.cloudflare.com`).
82     2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ?
83     "I'll use my own nameservers"*** ? paste the two Cloudflare nameservers ? Save.
Propagation:
84     minutes to 24 h. (Registration/renewal stays at GoDaddy; only DNS moves. Optional
later:
85     transfer the registration to Cloudflare Registrar for at-cost .guru renewals.)

```

```

86      3. Records (I manage after takeover): `app` ? Pages project; `api` ? `<tunnel-
id>.cfargotunnel.com`
87      (created automatically by `cloudflared`); apex redirect ? `app`.
88
89      ## 6. What I need to take over management
90
91      | Access | Scope | Used for |
92      |---|---|---|
93      | **Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS:Edit,
Account:Pages:Edit, Account:Access:Apps:Edit, Account:Tunnel:Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
94      | **GitHub repo** `khelsutra` (private) | Pages connects once (dashboard, you; build
root = `web/`) ? every push auto-deploys | I write the SPA in your Projects folder; commits
pushed = deployed |
95      | **On the box** (one-time, I hand you exact commands) | `cloudflared` installed as a
Windows service + `config.yml` | The tunnel itself |
96      | **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin,
Storage admin, Cloud SQL admin) | Control-plane deploys |
97
98      Day-to-day flow after setup: I build ? commit ? push ? Pages deploys the SPA
automatically;
99      API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never
live in the
100     repo; `.env` on the box + Cloudflare/GCP dashboards.
101
102     ## 7. Go-live security checklist (alpha)
103
104     Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate
105     limit + per-user upload quota ? `security.allowed_video_root` set ? upload size cap +
extension
106     allowlist ? **rotate the Gemini API key** (NEXT_STEPS housekeeping ? it was shared in
chat) ?
107     confirm no router port-forwards ? weekly backup: SQLite DB + `output/` ? GCS bucket
(rcclone
108     scheduled task; credits).
109
110     ## 8. Decisions needed from owner
111
112     1. Ratify the **hybrid recommendation** (CF free edge now, GCP credits at beta)? Or
force
113     single-vendor (DO / GCP) despite the cost notes above?
114     2. OK to move **DNS to Cloudflare** (registration stays at GoDaddy)?
115     3. Beta reel storage: **GCS (credits)** vs **R2 (zero egress)** vs **DO Spaces (1 TiB
included)** ?
116     can defer until beta Drop 2.
117     4. Approve the access grants in §6 so I can run the setup end-to-end.
118
119     *Ratified choices get recorded in `docs/archives/decisions/DECISIONS.md`.*
120

```

7. user (2026-06-20T08:59:35.146Z)

```

1      # Video locality ? where the bytes live (mental model + open questions)
2
3      **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4      the upload/storage problem, with the open questions named. **No implementation is
5      greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
6      **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7      `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
8      collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10     ## 0. The questions that triggered this (and the short answers)
11
12     ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow

```

```

13     assembles the full file under `security.upload_dir` (default `output/uploads`) on
14     our box, registers it by MD5, and nothing ever deletes it (retention policy is an
15     acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16     frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17     re-encoded chunks to Google's Files API.
18
19     ***"How would they upload a 10 GB file?"** Today they can't: `/api/upload/init`
20     rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21     under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22     whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23     video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
24     ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
25     miserable everywhere; expecting academies to manage S3/GCS buckets is a
26     non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
27
28     ***"Can we make it so there is NO upload?"** Mostly ? because of $1.
29
30     ## 1. The central insight: timestamps are the product; the video is dead weight
31
32     Walk the pipeline and ask what each stage actually consumes:
33
34     - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
35       **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
36     - The **detectors/featurizer** consume downsampled frames and trajectories.
37     - The only consumer of the original full-res bytes is the **highlight compiler**,
38       and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
39       original lives**.
40     - What the customer receives is `{rally intervals, report, reel}`. Intervals +
41       report are **kilobytes**. Given the intervals, the reel can be cut on the
42       machine that already has the original.
43
44     So: **any design that moves the 10 GB is moving the wrong thing.** The question
45     is never "how do we upload the video" but "what is the smallest artifact that
46     must cross the network for this tier of service."
47
48     ## 2. The locality ladder
49
50     From most to least bytes leaving the academy's Windows desktop. Sizes are rough
51     ratios for a 4K source ? measure on a real file before committing to any tier.
52
53     | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
54     |---|---|---|---|---|
55     | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
56     edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
57     files |
58     | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
59     client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
60     as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
61     manifest) | contract proven; client wrapper + server acceptance = new work |
62     | **L1 ? chunks/candidates upload** | only the ?1280px AI chunks (or just candidate
63     windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU);
64     server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's
65     own flow, relocated) |
66     | **L1-direct** | chunks go **straight from the client to Gemini** ? zero video bytes
67     through our server | ~0 through us | per-user key custody / metering / cost ceilings (OQ4) |
68     idea only |
69     | **L0 ? fully local** | nothing (calls to Google only if the Gemini path is used) | 0 |
70     the owned on-device model (roadmap north star) or a local GPU + our stack | not alpha-feasible;
71     the long-term answer |
72     | **L4 ? sneakernet** | a hard drive | 0 over network | a person (the field associate /
73     owner) | how the pilot actually works today |
74
75     **The honest "no upload" answer:** *strictly* no bytes leave only at L0 with the
76     owned model (and even L0 leaks ?1280px chunks to Google while the Gemini path is
77     the brain ? say that plainly in consent language, OQ3). The pragmatic near-term

```

65 answer is ****L2 or L1: don't eliminate the upload, shrink it 5?30x**** ? at which
66 point the existing 90 MB-part flow, the 4 GB cap, and a coffee-break wait are all
67 suddenly fine.

68

69 **## 3. Second-order effects worth noticing**

70

71 - ****L1/L2 make the serving box GPU-less for the Gemini tier**** ? ffmpeg + API
72 orchestration only. Hosting cost collapses; the GPU box stays for WASB/owned-
73 model work. (PLATFORM \$3's ComputeBackend split falls out naturally.)

74 - ****Storing the proxy instead of the original**** is ~10x less disk AND a smaller
75 privacy surface; the review UI (alpha screen A4) actually **prefers** streaming a
76 1080p proxy to seeking inside a 10 GB original.

77 - ****The correction loop survives**** at every tier ? L1: labels are intervals on a
78 timeline, and the proxy timeline IS the original timeline by construction
79 (the collector ADR-002 invariants: same fps, same frame count, parity-verified).

80 - ****video_id keying:**** the indexer keys by MD5 of the file it processed. At L2
81 that's the proxy's MD5; the proxy-manifest pattern (original_md5 ? proxy_md5)
82 already exists ? but the product needs ONE registered identity rule (OQ2).

83

84 **## 4. The frugality frame ? afford the bills by not burning GPU**

85

86 The runway constraint (owner, 2026-06-12): ****the bills stay personally affordable**
87 for a long time as long as we don't burn too much GPU.****** The locality ladder maps
88 directly onto that:

89

90 - ****The Gemini serving path costs ~\$0.30?0.60 per hour of video**** in API calls
91 (2026-06-10 quality table) and at L1/L2 needs ****no GPU at all**** ? a CPU box,
92 ffmpeg, and an API key. At pilot scale (a handful of academies, a few sessions
93 a week) that is single-digit dollars a month: affordable indefinitely.

94 - ****GPU spend is the thing to guard.**** WASB / owned-model inference and training
95 stay on the owner's already-paid-for local 2070S; rent nothing until the owned
96 model has a ship-gate-passing reason to be served. L2/L1 turn this from a
97 discipline into a default: **the serving box has no GPU to burn.**

98 - ****Storage follows the same logic:**** keep proxies, not originals (~10x less
99 disk); settle retention (OQ1) before volume arrives; the free edge tier stays
100 adequate precisely because L2 keeps payloads under the existing caps.

101 - ****Put an explicit cost ceiling per video**** (ties into OQ4): a 7-hour tape or a
102 runaway retry must not surprise-bill. The existing ops learnings (serial
103 uploads, `ai\_max\_workers=1`, bounded exponential retry) already point this way.

104

105 **## 5. The academy-visit dividend ? field work builds the product, not just the verdict**

106

107 The C13 visits aren't only a market read; run deliberately, each visit deposits
108 product assets:

109

110 - ****Phase-2 recordings (consented, adults-only) ARE the target corpus.**** The
111 owned model's ship target (0.66-to-beat) lives on exactly the messy multi-court
112 footage class these academies produce ? every show-back visit grows the
113 training set the model is starved for, at zero marginal engineering cost.

114 - ****The show-back is a live dress rehearsal of the product loop:**** footage comes
115 back on a drive (L4), we proxy + process + return a reel ? the same motion
116 L2 will automate. Every friction the associate hits there (file handoff,
117 proxy time, reel delivery) is a pre-paid product-backlog item; write it down.

118 - ****Demo reactions and improvement asks**** (captured verbatim on the answer
119 sheet) are the build backlog, collected by someone already standing in front
120 of the user.

121 - ****Connectivity notes from visits**** answer OQ8 with data instead of guesses.

122

123 **## 6. Open questions (the decision backlog this doc exists to hold)**

124

125 | # | Question | Why it matters / current lean |

126 |---|---|---|

127 | OQ1 | ****Retention policy per artifact class**** ? original (only L3 has it), proxy,
chunks, frames/trajectories, Gemini remote files, intervals/labels | intervals/labels are the
asset (keep forever, tiny); originals/proxies need an explicit keep-while-active + delete-on-

```

request rule; Gemini remote-file GC already in BACKLOG |
128   | OQ2 | **Which file's MD5 is the video's identity** at L2/L1 | lean: the processed
file's MD5 (proxy), with original_md5 carried in metadata ? the collector manifest pattern,
promoted to a product rule |
129   | OQ3 | **Consent language for the Gemini path** ? even "local" modes ship 1280px chunks
to Google | must be plain in the academy-facing terms; paid-tier no-training posture to be cited
|
130   | OQ4 | **Key custody for L1-direct** ? client calls Gemini with whose key? per-user
metering, abuse, cost ceiling per video | unsolved; this is what separates L1 from L1-direct |
131   | OQ5 | **Windows client packaging** ? portable ffmpeg + a signed tiny app vs a
PowerShell script for the pilot | pilot can be a script the associate/owner runs; product needs
an installer story (ffmpeg LGPL build = licensing-clean) |
132   | OQ6 | **What does the review UI play** at each tier | L2 streams the proxy (good); L1
has only chunks ? review degrades ? this quietly argues L2 over L1 as the default |
133   | OQ7 | **Reel delivery** ? server-compiled from proxy (shareable 1080p) vs intervals-
back + client cuts full-res | could be both: proxy reel by default, "premium full-res" cut
locally |
134   | OQ8 | **Real academy uplink numbers** ? the L2-vs-L3 case rests on bandwidth
assumptions | C13 visits can note connectivity informally; don't redesign around guesses |
135   | OQ9 | **Recurring volume** ? an academy recording daily generates 10 GB/day; even L2
accumulates | retention (OQ1) + maybe future delta thinking; not an alpha problem |
136   | OQ10 | **If L3 must survive** (users who refuse any client-side step) | needs
resumable uploads + a paid edge tier + a raised cap; treat as the exception path, not the design
center |
137
138   ## 7. Where this lands (recommendation, not a greenlight)
139
140   **Default the alpha to L2 (proxy upload):** a one-command client prep step built
141   on the *already-verified* proxy contract, uploads that fit comfortably inside
142   today's chunked endpoint and 4 GB cap, a GPU-less serving path (the affordability
143   requirement, $4, satisfied by construction), a review UI that streams the proxy,
144   and intervals that map 1:1 onto the customer's original.
145   **L1/L1-direct is the fast-follow** once OQ4 has an answer; **L0 is the owned-
146   model milestone**, not an alpha decision. The field pilot keeps running on L4 ?
147   a person with a hard drive is a fine storage backend for n=3 academies.
148

```

8. assistant (2026-06-20T08:59:37.177Z)

Now let me read the remaining docs and deploy scripts.

9. user (2026-06-20T08:59:38.542Z)

```

1      # Cross-Platform Desktop App ? "Plug in your card, get your highlights"
2
3      > **Status:** `DRAFT` ? `owner-gated` (scoping doc only; **nothing is built**) .
**Owner:** avidullu . **Created:** 2026-06-18 . **Last updated:** 2026-06-18
4      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once the owner
greenlights P0).
6      > **Relation to existing docs:** **peer-of**
[PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) (this is the **`LocalDesktop`
ComputeBackend** made into a product) . **realizes the L0 tier** of
[VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md) . **consumes**
[OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) (the smaller/export-
clean model is one answer to the compute crux) . bound by
[COMMERCIALIZATION.md](COMMERCIALIZATION.md) licensing guardrails.
7      > **Honesty note:** every claim is tagged `[verified]` (checked against code/measured),
`[researched]` (needs sign-off), or `[design]` (proposed). **The entire build is `[design]`** ?
this doc scopes; it does not ship. Not legal advice.
8
9      ---
10

```

```

11      ## 0. TL;DR
12      Package the mature local pipeline (WASB shuttle detector + improved windowing + ffmpeg
stitch, already runnable fully-offline via [`tools/generate_highlights.py --skip-ai-
handoff`](../tools/generate_highlights.py)) as a cross-platform desktop app for non-
technical recreational players: plug in the camera's USB/SD card, click once, get back only
the highlight reels ? fully on-device, no upload, no copy of the GBs of 4K originals.
This is the product surface that fits the "local, cheap, efficient" thesis and the L0 ("nothing
leaves the machine") tier of `VIDEO_LOCALITY_MODEL.md`.
13
14      The make-or-break unknown is compute on a typical enthusiast laptop ? which often
has no discrete GPU. WASB runs at ~3.4x real-time on a laptop RTX 2070 Super `[verified,
measured]`; on CPU it is far slower, and Apple Silicon has no CUDA at all. P0 is a
feasibility spike that measures CPU and Apple-MPS throughput for a ~10-min clip before any app
is built. If CPU-only is unacceptable on commodity hardware, the path needs a smaller/faster
owned model (ties `OWNED_MODEL_IMPLEMENTATION_PLAN`) or a tiered "fast-if-GPU, slow-but-works-
if-CPU" design. Everything past P0 is gated on that result.
15
16      ## 1. Problem & goal
17
18      Why now / the user. Target users are recreational sports enthusiasts, not tech-
savvy. Their match videos are huge (GBs of 4K). The cloud-upload product surface fails
them on every axis: slow on a typical Indian uplink (`VIDEO_LOCALITY_MODEL.md` S0: 10 GB ? ~1 h
at 20 Mbps, ~4.5 h at 5 Mbps), bandwidth-costly, privacy-fraught, and it bloats both our storage
and their drives. The thing the customer actually wants back is kilobytes of timestamps ? a
highlight reel `(VIDEO_LOCALITY_MODEL.md` S1: "timestamps are the product; the video is dead
weight")`.
19
20      The concrete outcome ("good" looks like this):
21      1. User plugs the GoPro/phone SD card or camera USB into their laptop.
22      2. The app detects the card, lists the match videos on it.
23      3. User clicks Generate Highlights.
24      4. The app makes an on-device proxy, runs WASB detection + windowing + stitches the reel
? 100% locally.
25      5. The reel(s) land in a user-chosen folder. The originals are never copied off the
card and never bloat any drive.
26
27      No account, no upload, no cloud dependency in the default path. Gemini stays an
optional "better brain if you want it" toggle ? never required, never bundled as a hard
dependency, never a training source (`COMMERCIALIZATION` C4).
28
29      Read to get current: `VIDEO_LOCALITY_MODEL.md` (the L0/locality framing + Q05
packaging seed), `PLATFORM_ARCHITECTURE.md` S3.2 (the ComputeBackend seam),
[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py) (the detector ABC
that already anticipates a Linux-native runner), `tools/generate_highlights.py` (the local
entrypoint this app wraps).
30
31      ## 2. Decisions locked
32
33      | # | Decision | Source / date | Implication |
34      |---|-----|-----|-----|
35      | D1 | Default path is 100% on-device, no upload, no Gemini. The reel is produced
from the local pipeline (`--skip-ai-handoff`). | This doc . 2026-06-18; matches
`VIDEO_LOCALITY_MODEL` L0 | The app must never require a network call. Gemini is an opt-in
toggle only. |
36      | D2 | Never copy the originals. The app reads source video in place (off the card /
a chosen folder) and writes only reels + a tiny report to a user folder. | This doc; mirrors
`generate_highlights.py` non-destructive `_atomic_publish` discipline | No "import library" that
duplicates 10 GB files. Detection runs on a proxy; stitch reads the original in place. |
37      | D3 | Drop the WSL2 dependency; run the detector natively per-OS. | This doc;
enabled by the `DetectorRunner` ABC `[verified]` | Windows-native (no WSL), Linux+NVIDIA (CUDA
directly), macOS (MPS or CPU). The WSL adapter stays for the current dev box but is not the
shipped path. |
38      | D4 | Commercial-clean licensing for everything bundled: MIT / Apache-2.0 / BSD
only (code, weights). ffmpeg ships as an LGPL build (no GPL-only encoders like x264). |
`COMMERCIALIZATION` (ratified 2026-06-09); `VIDEO_LOCALITY_MODEL` Q05 (ffmpeg LGPL build =

```



```

licensing-clean") | WASB-SBDT weights are MIT `[verified]` (see `tracknet-wasb-wsl2-decision`).
The stitch must use openh264 (BSD) or OS hardware encoders, not statically-linked x264. See
$3.4. |
39 | D5 | Reuse the existing pipeline verbatim; add only the shell + a native runner +
device-IO.** | `backend/pipeline/detectors/base.py` reuse contract `[verified]` | No re-
architecture of windowing/stitch. New code = native runner, USB/SD detection, app shell,
packaging. |
40 | D6 | Feasibility is gated on P0. No app shell, no packaging work starts until the
compute spike returns a verdict. | This doc; `decision-rigor-norm` (NUMBERS INFORM, FOUNDATIONS
DECIDE) | P0 is the only un-gated deliverable; P1?P3 are owner-greenlit after P0 numbers. |
41
42 ## 3. Foundation ? research / prior art `[design]` / `[researched]`
43
44 ### 3.1 The compute reality (the crux) ? what we know vs must measure
45 - WASB on a discrete laptop GPU is acceptable `[verified, measured]`: on the owner's
RTX 2070 Super Max-Q (8 GB), WASB inference is the pipeline driver at ~3.2?3.4x real-
time (?3.4 GPU-min per footage-min, ~0.055 s/frame), peak VRAM ~5.7?6.0 GB; proxy (NVENC)
~0.2x RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end ?5x clip length. Source: `current-
state.md` `COST_SCALING.md` checkpoint (6 Boxhill clips, 2026-06-18).
46 - The typical enthusiast laptop has NO discrete GPU. A 10-min 4K clip at 3.4x RT is
~34 GPU-min with a 2070S. On integrated graphics / CPU-only the wall-clock is unknown and
could be 10x+ worse ? possibly hours per clip, which would kill the one-click promise.
47 - Apple Silicon has no CUDA. WASB/TrackNet weights are PyTorch; PyTorch MPS (Metal)
backend may run them, but op-coverage and speed on the WASB model are unverified
`[design]`. CPU fallback on Apple is also unverified.
48 - Owned-model tie-in. `OWNED_MODEL_IMPLEMENTATION_PLAN.md` already states the on-
device answer: train in Python ? export across the boundary (ONNX / Core ML / ExecuTorch /
GGUF) ? a thin device shell runs the artifact with zero Python*, with the standing constraint to
train with export-clean ops. A smaller, exported, quantized shuttle detector is a credible
answer if the CPU/MPS numbers on the current WASB weights are bad. This makes P0 partly a
referendum on whether the desktop product needs the owned-model track.**
49
50 ### 3.2 The de-WSL native runner is already designed-for `[verified]`
51 The detector layer was deliberately built to make this port low-regret.
[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py) documents,
verbatim, an "Optional Linux-native path (the main de-risk goal)": a
`LinuxNativeRunner`/`ONNXRunner` that "Skip[s] all wsl / _stage / conda activate, load[s]
weights locally (torch or onnx)? write[s] the exact same CSV format so
`trajectory_to_action_windows` works verbatim." The WSL plumbing lives in a separate
`WslCommandMixin`, explicitly so a native runner "must not inherit WSL plumbing." The actual
inference math is already isolated in
[`wasb_infer.py`](../backend/pipeline/detectors/wasb_infer.py) (torch + the WASB-SBDT modules,
emitting `Frame,Visibility,X,Y`). So the port is: lift `wasb_infer.py` out of WSL, load
weights with native torch/onnxruntime per-OS, keep the identical CSV contract. The windowing
brain (`trajectory_to_action_windows`) and stitch are untouched.
52
53 ### 3.3 Compute backends to evaluate in P0
54 | Path | OS | Toolkit | Status |
55 |---|---|---|---|
56 | CUDA-native | Windows-native, Linux+NVIDIA | torch+CUDA (no WSL) | proven in WSL
`[verified]`; native = port |
57 | Apple MPS | macOS (Apple Silicon) | torch MPS / Core ML export | unverified ? P0
must measure** |
58 | CPU | all (the no-GPU laptop) | torch-CPU / onnxruntime (often fastest on CPU) |
unverified ? P0 must measure; the gating number** |
59 | Smaller/exported model | all | ONNX/Core ML/ExecuTorch, quantized | deferred to owned-
model track; P0 informs whether it's needed |
60
61 ### 3.4 App-shell prior art & licensing landscape `[researched]`
62 - App shell options: Tauri (Rust + system webview; ~3?10 MB binaries;
MIT/Apache) vs Electron (bundled Chromium; ~100?150 MB; MIT) vs CLI + installer (wrap
`generate_highlights.py`; smallest effort). All shell licenses are clean.
63 - Backend packaging: the Python/FastAPI backend bundles via PyInstaller or a
frozen venv as a local "sidecar." Packaging was already modernized ? PEP 621 `pyproject.toml`
+ an indexer console entry point shipped (`DEFERRED_HARDENING_PLAN` #7, DONE, PR #167)

```

```

`[verified]` ? so a CLI MVP has a real head start.
64 - **ffmpeg licensing (a genuine landmine):** ffmpeg is LGPL/GPL. The current stitch path
uses **libx264 (GPL)**. Redistributing that inside an installer would impose GPL on the bundle.
**Mitigation (D4):** ship an **LGPL ffmpeg build** using **openh264 (BSD)** or the OS hardware
encoder (NVENC / Apple **VideoToolbox** / Linux **VAAPI**) ? the desktop app should prefer
hardware H.264 anyway (faster, no CPU stitch tax).
65 - **Weights:** WASB-SBDT = MIT `[verified]` (`tracknet-wasb-wsl2-decision`),
redistributable.
66
67 ## 4. Design / architecture `[design]`
68
69 ### 4.1 The one-click flow (and what code each step reuses)
70 ```
71 [card inserted]
72 ? (NEW) per-OS device-insert watcher ? mount path
73 ?
74 [list videos on the card] ? glob video exts; reuse _VIDEO_EXTS from
generate_highlights.py
75 ?
76 [make on-device proxy] ? ffmpeg downscale (hardware encoder); reuse the proxy +
--detect-from split (#205)
77 ?
78 [detect: WASB native] ? (NEW) NativeWasbRunner (DetectorRunner subclass) ? NO WSL;
same CSV contract
79 ?
80 [windowing ? rallies] ? REUSE trajectory_to_action_windows + improved milestone
windowing (R1: cap=6 + R4)
81 ?
82 [stitch reel from ORIGINAL] ? REUSE VideoCompiler
(backend/pipeline/stitcher/compiler.py); read original in place
83 ?
84 [export reel + tiny report to a user folder] ? reuse _atomic_publish non-destructive
write; NEVER copy originals
85 ```
86 The detector runs on a **proxy**; the stitch reads the **original on the card** (the
`--detect-from` split, already shipped `[verified]` in `generate_highlights.py`). Rally
timestamps transfer 1:1 because the proxy shares the timeline (the collector frame-identical
proxy contract, `VIDEO_LOCALITY_MODEL` $1).
87
88 ### 4.2 Reuse map ? what's NEW vs REUSED
89 | Concern | Status | Code |
90 |---|---|---|
91 | Trajectory?rally windowing | **REUSE** `[verified]` |
`tracknet_runner.trajectory_to_action_windows` (model-agnostic staticmethod) |
92 | Stitch / reel compile | **REUSE** `[verified]` |
`backend/pipeline/stitcher/compiler.py::VideoCompiler` (pure ffmpeg) |
93 | Local orchestration (proxy?detect?stitch, non-destructive, resumable, `--skip-ai-
handoff`, `--detect-from`) | **REUSE** `[verified]` | `tools/generate_highlights.py` |
94 | Detector runner contract | **REUSE** `[verified]` |
`backend/pipeline/detectors/base.py::DetectorRunner` ABC |
95 | WASB inference math | **REUSE** (lift out of WSL) `[verified]` |
`backend/pipeline/detectors/wasb_infer.py` |
96 | **Native** (de-WSL) WASB runner, per-OS** | **NEW** `[design]` | `NativeWasbRunner` ?
sibling of `wasb_runner.py`, no `WslCommandMixin` |
97 | **USB/SD** insert detection + video listing** | **NEW** `[design]` | per-OS: Win
WMI/`WM_DEVICECHANGE`, macOS DiskArbitration, Linux udev/udisks2 |
98 | **App shell / one-click UI** | **NEW** `[design]` | Tauri vs Electron vs CLI ($3.4; P2
decides) |
99 | **Packaging + code-signing per OS** | **NEW** `[design]` | PyInstaller/MSIX (Win),
.dmg+notarize (mac), AppImage/Flatpak (Linux) ? P3 |
100
101 ### 4.3 Where this sits in the platform architecture
102 This **is** the `local` / `LocalDesktop` **ComputeBackend** of
`PLATFORM_ARCHITECTURE.md` $3.2, realized as a shippable product instead of a server worker. The
control-plane/data-plane split degenerates cleanly to "all on one machine." Nothing here

```

forecloses the hosted SaaS path ? it's the privacy-max end of the same spectrum, and a genuine differentiator (no competitor offers true on-device compute except SwingVision, which proves it's commercially viable ? `PLATFORM\_ARCHITECTURE` \$intro).

103

104 ### 4.4 App-shell recommendation (to confirm at P2)

105 Lean: ****start with a CLI + installer MVP**** (wraps the existing
`indexer`/`generate\_highlights.py` entry point ? fastest way to validate packaging, signing,
USB-detect, and the native runner end-to-end on all three OSes), then ****graduate to a Tauri
GUI**** for the non-technical user (tiny download fits the "huge-video laptops, slim app" frame;
the Python backend runs as a sidecar the Tauri front-end calls over localhost). Electron is the
fallback if webview inconsistencies bite. This mirrors the repo's "simplest thing first,
graduate on evidence" bias.

106

107 ## 5. Risk table `[design]`

108 | Risk | Severity | Mitigation / where decided |

109 |---|---|---|

110 | ****CPU/MPS too slow on a no-GPU laptop**** (kills the one-click promise) | ****make-or-
break**** | ****P0 spike measures it before anything is built**** (\$6, \$7-P0). Fallbacks: tiered
fast/slow UX, or the owned smaller model. |

111 | ffmpeg GPL contamination via libx264 in the installer | high (legal) | D4: LGPL build
+ openh264/hardware encoders only. |

112 | Code-signing friction (mac notarization mandatory; Win SmartScreen without EV cert) |
medium | P3 scopes certs/notarization per OS; budget the Apple Developer Program + a Windows
cert. |

113 | Privacy regressions (the on-device promise is the **selling point**) | medium | D1/D2:
no upload, no original-copy, no required network. Make the "nothing leaves your machine"
guarantee testable (assert no outbound calls in default mode). |

114 | Per-OS device-insert flakiness / removable-media edge cases (card pulled mid-run) |
medium | P1/P2 handle gracefully; never write to the card; fail safe. |

115 | Support burden for non-technical users (drivers, "it didn't detect my card") | medium
| favors a polished GUI + clear errors (reuse the actionable-error discipline already in the WSL
adapter). |

116

117 ## 6. Honest limits ? what this does NOT do

118 - ****A DRAFT/DONE status here proves nothing about runtime feasibility.**** The doc is a
scope; only the P0 numbers decide whether the product is buildable on commodity hardware.

119 - ****If CPU-only is too slow on a typical laptop, the simple "wrap the pipeline" path
does not exist**** ? it becomes "ship a smaller owned model" (a much larger effort, ties
`OWNED\_MODEL\_IMPLEMENTATION\_PLAN`) or "GPU-tier only" (shrinks the addressable market). The doc
names this honestly rather than assuming the wrap is free.

120 - ****No accuracy claim changes.**** The desktop app inherits exactly the current rally-
detection quality (R1 milestone: precision is still the open gap per
`RALLY\_DETECTION\_GAP\_CLOSURE.md`); packaging does not improve detection.

121 - ****Gemini-quality reels need the optional online toggle.**** The fully-local default uses
`--skip-ai-handoff`; users wanting the Gemini brain accept that ?1280px chunks leave the machine
(consent language, `VIDEO\_LOCALITY\_MODEL` OQ3).

122 - ****macOS support is contingent on P0's MPS/CPU verdict**** ? it is not assumed.

123

124 ## 7. Deliverables & progress tracker ? ****source of truth****

125 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
****Everything below is owner-gated; only this doc (the scoping PR) is in flight today.****

126

127 | ID | Deliverable | Depends on | Gated? | Status | PR |

128 |---|-----|-----|-----|-----|-----|

129 | ****D0**** | ****This scoping doc**** (`DESKTOP\_APP\_PLAN.md` + README index) | ? | No | ? |
(this PR) |

130 | ****P0**** | ****Compute-feasibility spike (make-or-break).**** Measure WASB throughput for a
~10-min clip on: (a) CPU-only (torch-CPU & onnxruntime), (b) Apple MPS, (c) confirm CUDA-native
(no WSL). Output: a wall-clock table + a go/no-go verdict + "do we need a smaller model?"
recommendation. ****No app code.**** | D0 | ****Owner greenlight**** | ? | ? |

131 | ****P1**** | ****De-WSL native detector port**** (`NativeWasbRunner`, per-OS): CUDA-native
(Win/Linux), MPS/CPU fallback, identical `Frame,Visibility,X,Y` CSV; reuse windowing verbatim;
healthcheck per backend. | P0 verdict | Yes | ? | ? |

132 | ****P2**** | ****Desktop app shell + one-click flow****: USB/SD detect ? list ? proxy ? native

```

detect ? stitch ? export; shell choice (CLI MVP ? Tauri) confirmed; "no upload / no original-
copy" enforced & testable. | P1 | Yes | ? | ? |
133 | **P3** | **Packaging + code-signing installers** per OS (Win MSIX/Authenticode, mac
.dmg + Developer-ID notarization, Linux AppImage/Flatpak); LGPL ffmpeg bundling; MIT/Apache/BSD
audit of the bundle. | P2 | Yes | ? | ? |
134
135 ## 8. Open questions ? owner / external
136 **Blocking gates (owner decides before the dependent phase):**
137 1. **(gates P1?P3) Greenlight P0?** ? the spike is cheap and decides everything.
Recommend: yes, run P0 next.
138 2. **(gates P1) If CPU is too slow, which fork?** tiered GPU-fast/CPU-slow UX · invest
in a smaller exported owned model (`OWNED_MODEL_IMPLEMENTATION_PLAN`) · GPU-tier-only product
(narrower market).
139 3. **(gates P2) App shell:** confirm CLI-MVP-then-Tauri, or go straight to a GUI /
Electron.
140 4. **(gates P3) Distribution & signing budget:** Apple Developer Program (mandatory for
notarization), a Windows code-signing cert (EV for clean SmartScreen) ? owner cost decision.
141
142 **Nice-to-knows:**
143 5. OS coverage priority for v1 (Windows-first? given the owner's box and the academy
beachhead).
144 6. Auto-update strategy (Tauri updater / Sparkle / none for the pilot).
145 7. Does the desktop app reuse the optional Gemini toggle at all, or stay strictly local
for v1?
146
147 ## 9. Definition of done
148 This subproject is DONE (? flip Status, archive) when:
149 - [ ] **P0** produced a measured throughput table (CPU / MPS / CUDA-native) and an
owner-accepted go/no-go verdict (incl. whether a smaller model is required). *(P0 alone is a
valid stopping point if the verdict is "not feasible without the owned model" ? then this doc
hands off to `OWNED_MODEL_IMPLEMENTATION_PLAN`.)*
150 - [ ] **P1** the native (de-WSL) detector runs on ?1 GPU OS and ?1 CPU/MPS path,
producing byte-compatible trajectories (regression vs the WSL runner on a shared clip).
151 - [ ] **P2** a non-technical user can insert a card and get a reel with no terminal, no
upload, and no copied original ? verified end-to-end on ?1 OS.
152 - [ ] **P3** signed/notarized installers exist for the greenlit OSes; the bundle passes
an MIT/Apache/BSD (+ LGPL-ffmpeg) license audit.
153 - [ ] $7 rows all ? and the headline feasibility verdict is recorded for posterity.
154
155 ## 10. References
156 **Internal code anchors `[verified]`:**
[tools/generate_highlights.py](../tools/generate_highlights.py) (local orchestration; `--skip-
ai-handoff`, `--wasb-stream`, `--detect-from`, `_WSL_DETECTORS`, `_atomic_publish`) ·
[backend/pipeline/detectors/base.py](../backend/pipeline/detectors/base.py) (`DetectorRunner`
ABC + the documented Linux-native de-risk path + `WslCommandMixin`) ·
[backend/pipeline/detectors/wasb_runner.py](../backend/pipeline/detectors/wasb_runner.py) (the
WSL adapter to replace) ·
[backend/pipeline/detectors/wasb_infer.py](../backend/pipeline/detectors/wasb_infer.py) (the
torch inference core to lift native) · [backend/pipeline/detectors/tracknet_runner.py](../back
end/pipeline/detectors/tracknet_runner.py) (shared `trajectory_to_action_windows`) ·
[backend/pipeline/segmenters/wasb_hybrid.py](../backend/pipeline/segmenters/wasb_hybrid.py) ·
`backend/pipeline/stitcher/compiler.py::VideoCompiler` · `pyproject.toml` + `indexer` console
entry (#167).
157 **Internal docs:** [VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md) (L0 fully-local
tier; Q05 packaging) · [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) §3.2
(ComputeBackend seam) ·
[OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) (export-clean on-
device model) · [COMMERCIALIZATION.md](COMMERCIALIZATION.md) (C4 paid-LLM no-train;
MIT/Apache/BSD-only) · [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) (#7 packaging,
DONE) · [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md) (the WSL setup being superseded for
shipped builds).
158 **Measured source:** `current-state.md` 2026-06-18 `COST_SCALING.md` checkpoint (WASB
~3.4× RT on RTX 2070 Super Max-Q, peak VRAM ~5.7?6.0 GB).
159 **External (to validate at P0/P2/P3):** PyTorch MPS backend · onnxruntime
(CPU/CoreML/DirectML execution providers) · Tauri vs Electron sizing · ffmpeg LGPL build /

```

```
160
161 ---
162
163 ### Changelog
164 - `2026-06-18` ? Created. Scoping doc for the cross-platform desktop app + the de-WSL
native-compute port; all `[design]`. $7 P0 (compute-feasibility spike) named the make-or-break
gate. Owner-gated; nothing built.
165
```

10. user (2026-06-20T08:59:38.571Z)

```
1      # Platform Architecture ? Multi-User "Assistant Coach"
2
3      **Status:** ? Strategy / research ? for review. No code yet.
4      **Date:** 2026-06-07 · **Owner:** avidullu
5
6      > ? **Superseded where it disagrees with
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
7      > (2026-06-09, authoritative).** Specifically: (1) licensing is **MIT/Apache-2.0/BSD**
(not "Apache-2.0 only");
8      > (2) base model is **Qwen3-VL (Apache)** primary / InternVL3 (MIT) fallback ? **Gemma-4
is AMBER/bench-only**, not a
9      > cleared base; (3) the owned VLM is **not** the per-turn "conversational brain" ?
extraction (TAL head ? later VLM)
10     > feeds a per-video **event store + text-to-SQL** query layer ("extract once, query
many"); the VLM-answers-each-turn
11     > design is rejected. The $5A model/base/QA-harness detail below predates that and is
kept for context only.
12
13     **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
14     "assistant coach" platform**: users upload videos, get highlights **stored and
15     browsable by history**, and choose **where their data lives** (Google Drive / cloud
16     bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
17     principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
18     execution/compute, the detection+highlight pipeline, and data-serving ? behind small
19     **pluggable interfaces**, reusing the registry pattern the codebase already uses for
20     segmenters/providers/validators/sports/annotations.
21
22     **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
23     not just a clipper ? but the strategy is **phased and unambiguous about scope**:
24     - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
25     history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
26     *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
27     - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
28     **owned video-QA model** ($5A; $6 P4?P5).
29     - **North-star vision (strictly future ? not* v1):** **gait / biomechanics, per-athlete
30     identifying templates, and deep coaching feedback under a supervising coach** ($6 P6).
31     Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
32     bar ($7).
33
34     The buyer is the **academy/coach** (beachhead in
35     [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
36     consequences ripple through the design: (1) the data model keeps an **optional coach ?
37     athlete overlay over a generic core** ($4.1) ? not baked in; (2) gait/biometrics is
38     **special-category data**, gated and deferred ($7).
39
40     > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
41     > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
42     > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
43     > your storage, your compute"** is a genuine differentiator for privacy-sensitive
44     > academies ? and SwingVision proves user-side compute is commercially viable. All four
```

```

45 > share the same **teams ? athletes ? clips** data model, which we adopt (§4).
46
47 > Scope: this is a **direction-setting doc**. It does not start implementation; the
48 > first infra slice (an async job model) is the on-hold #16 in
49 > [ `DEFERRED_HARDENING_PLAN.md` ](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
50 > from [ `COMMERCIALIZATION.md` ](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
51 > foundations) and `archives/MONETIZATION_AUDIT.md`.
52
53 ## 1. Current state & gaps (what we're evolving from)
54
55 | Concern | Today | Gap for multi-user |
56 |---|---|---|
57 | Identity / tenancy | none | no users, orgs, auth, or isolation |
58 | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
59 | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
60 | Execution | **async job model shipped** ? `/api/process` returns **202 + job_id**, a
single-worker `ThreadPoolExecutor` drains the `jobs` table, `GET /api/jobs/{id}` polls status |
single-tenant + local-only; can't dispatch to a user's VM/GPU yet |
61 | Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation
registries) | reuse as-is ? this layer is the model for the rest |
62 | API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs
auth, per-tenant scoping, async job + history endpoints |
63
64 Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
65 migration ladder, **v5** ? `jobs` table + per-user `user_models` already landed),
66 `backend/main.py` (endpoints + the async job worker ? `/api/process` ? 202 + job_id,
67 `GET /api/jobs/{id}`), the registry pattern in
68 `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
`backend/providers/base.py`.
69
70 ## 2. Target layered architecture
71
72 Decouple into layers with a clean **control-plane vs data-plane** split ? the
73 orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live
74 elsewhere (even in the user's own cloud).
75
76 ...
77
78 CONTROL ? ??????????????????????????????????????????????????????????????
79 PLANE ? Presentation / API (auth, per-tenant) ?
80 ? Identity & Tenancy (org?coach?athlete?user) ?
81 ? Orchestration / Jobs (submit?status?result) ?
82 ??????????????????????????????????????????????????????????????
83 ? dispatch ? metadata
84
85 DATA ? ??????????????????????????????????????????????????????????????
86 PLANE ? Execution / Compute ? ? Data / Serving ?
87 ? (local | BYO-VM | ? ? (per-tenant DB: ?
88 ? hosted GPU) ? ? videos, segments, ?
89 ? runs the Pipeline: ? ? highlights, jobs, ?
90 ? detect ? highlight ? ? history) ?
91 ??????????????????????????????????????????????????????????????
92 ? read input / write artifacts
93 ??????????????????????????????????????????????????????????????
94 ? Storage (local upload | Google Drive | S3 | ?
95 ? GCS | Azure) ? raw video + highlight outputs ?
96 ??????????????????????????????????????????????????????????????
97
98 The two **new pluggable seams** are **Storage** and **Compute/Execution** ; everything
99 else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
100 **model/training layer** ? the owned generative video-QA model (§5A) ? sits atop these,
fed by Storage (training corpus), Data (labels), and Compute (training + inference), and

```

```

101 is delivered through the existing **AI-provider registry with a paid fallback**.
102
103 ## 3. Pluggable abstractions (extend the existing registry pattern)
104
105 The codebase already proves the model: an `ABC` + a `Registry` singleton with
106 `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
107 the same shape so they're plug-and-play and testable in isolation.
108
109 ### 3.1 `StorageBackend` registry
110 Replace the bare `video_path: str` that flows through the pipeline with a
111 **`StorageRef`** (a backend id + a key/URI), resolved by a backend.
112
113 ```python
114 class StorageBackend(ABC):
115     name: str
116     def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
117     def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
118     def upload(self, local_path: str, ref: StorageRef): ... # persist
119     def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
120     def stat(self, ref: StorageRef) -> StorageStat: ...
121     def list(self, prefix: StorageRef) -> list[StorageRef]: ...
122 ```
123
124 - **Recommended implementation:** build the backend over **`fsspec`** (one matrix ?
125 local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
126 Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4
127 byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
128 add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
129 - **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
130 the tenant registers their own bucket/Drive; raw video **never leaves their account**.
131 - **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
132 IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned
133 URLs** so raw video moves **browser ? the user's bucket directly, never through our
134 servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academics, a
135 **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
136 - **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while
137 the LLM provider can take a **signed URL**. The backend exposes both `download_to_local`
138 (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
139 caring **where* the bytes live.
140 - **Migration path:** introduce `StorageRef` alongside the current local path; a `local`
141 backend wraps today's behavior so nothing breaks; adapters land incrementally.
142
143 ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
144 Where the pipeline actually runs for a given job.
145
146 - **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).
147 - **`hosted`** ? our GPU VM pool (the SaaS default).
148 - **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their
149 own VM/GPU. **Critical security property:** the worker **pulls* jobs scoped to its tenant
150 from the control plane, pulls input from the tenant's **own** storage, runs the
151 pipeline, and writes artifacts + results back ? so **the orchestrator never holds the
152 user's cloud credentials** and needs **no inbound access** to their network. This
153 mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s
154 outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most
155 credential-safe v1** for BYO-compute.
156 - For users who want the platform to **provision GPUs in their *own* cloud account on
157 demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot**
158 (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
159 **Modal** is the managed fallback for users wanting zero infra (accepting that raw
160 video then flows through Modal).
161 - **?? Operational / support burden (a real adoption risk).** For the primary beachhead

```

?

162 Indian badminton ****academies/coaches**** ? asking them to ***run and maintain a worker on**
 163 **their own VM/GPU*** is a meaningful ongoing load (updates, monitoring, firewall/NAT, the
 164 "it stopped after our IT change" ticket). So: ****default academy users to `hosted`****,
 and
 165 treat **`byo\_worker`/SkyPilot** as a ****power-user / enterprise / privacy-sensitive tier****,
 166 not the academy default. If/when BYO ships, plan for ****auto-updating, self-healing**
 167 **agents + health heartbeats + clear worker status in the UI****, and budget the support
 cost.
 168 - ****Middle tier to consider ? "zero-infra BYO":**** a managed-but-isolated path where the
 169 ****storage**** stays in the user's bucket (signed URLs) while compute runs on ****our****
 GPUs
 170 in a ****strictly isolated, per-tenant, ephemeral sandbox**** (nothing persists our side).
 171 This gives the BYO ***privacy*** benefit (raw video never persists on our infra)
****without****
 172 the user running a worker ? a pragmatic default between full-hosted and full-BYO.
 173
 174 The pipeline code is unchanged; only ***where it's invoked*** and ***how it gets bytes*** vary.
 175
 176 **### 3.3 Reuse, don't reinvent**
 177 The existing **`segmenter` / `provider` / `validator` / `sport` / `annotation`** registries
 178 are the proven precedent and stay as the ****pipeline**** layer. The **`annotation`** registry +
 179 **`candidate\_segments`** table are also the natural seam for the ****coach-correction / golden**
 180 **-set**** loop in the assistant-coach phase (§6, P4).
 181
 182 **## 4. Multi-tenant data model & serving layer**
 183
 184 **### 4.1 Entities ? generic core, with the coach model as an optional overlay**
 185 The foundational schema is deliberately ****domain-neutral**** so we don't overfit to the
 186 coach mindset ? the coach/athlete framing is a ****market-fit hypothesis that will**
 iterate**
 187 (see C13 / PMF), not a load-bearing assumption. ****Generic core:****
 188 ```
 189 User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
 190 ???< PlaySegment ??< Event
 191 Video / Highlight ??< Job (processing record: status, compute backend, timings)
 192 ```
 193 - ****Generic & re-targetable:**** a **`VideoCollection`** groups a user's videos (a season, an
 194 event, a "folder") with no assumption of a coach or academy ? a power user, a coach,
 or a
 195 parent are all just **`User`**'s with collections.
 196 - ****Highlights are first-class**** (today compile only writes a file): a **`highlights`**
 table
 197 (segment ids + output **`StorageRef`** + params + created_at) powers "see past videos and
 198 highlights by history," and ****`HighlightMetadata`**** carries the ****rich semantic tags****
 199 (shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings /
 notes.
 200 - ****`video\_id`** stays the content MD5** (dedup within a tenant), scoped by **`tenant\_id`**.
 201 - ****Coach overlay (optional, additive):**** for the academy/coach GTM, layer
 202 **`Org/Academy ? Coach ? Athlete`** *on top* (e.g. a **`Membership`/role** table + an
`athlete\_id`
 203 tag on collections/videos) ****without**** baking it into the core tables ? so the schema
 204 stays usable for solo power users and the coach model can evolve, or be dropped, as
 PMF
 205 dictates.
 206 - ****History/serving API:**** list collections/videos/highlights per user (and, with the
 207 overlay, by athlete/coach/date); ****query highlights by `HighlightMetadata`**** (e.g.
 shot
 208 type / outcome); re-stitch from stored segments without reprocessing.
 209
 210 **### 4.2 Tenancy isolation ? options & recommendation**
 211 | Model | Isolation | Ops cost | Best for |
 212 |---|---|---|---|
 213 | ****Shared DB + `tenant\_id`**** (row-level, ideally Postgres ****RLS****) | logical | low |
****Hosted SaaS, many small tenants**** ? ***recommended default*** |


```

214 | Schema-per-tenant | medium | medium | tens?hundreds of mid tenants |
215 | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**self-hosted single-academy**, or high-isolation enterprise |
216
217 **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
218 `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
219 single-academy** deployment (it reuses today's code almost verbatim). Either way,
220 **abstract DB access behind a repository layer now** so the engine is swappable, and
221 graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.
222
223 ## 5. Orchestration / async jobs
224
225 This is the serving layer that makes multi-user real. It is the **multi-user superset of
226 [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ?
build
227 that first, single-tenant, then generalize.
228
229 - **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),
230 status (queued|running|done|failed), progress, compute_backend, result_ref, error,
231 timestamps`. `/api/process` returns **202 + job_id** ; `/api/jobs/{id}` polls.
232 - **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
233 content-hash discipline already in `compute_video_id`.
234 - **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
235 for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
236 writes artifacts + DB rows. **Observability reports** (already per-video) become the
237 job artifact surfaced in history.
238 - **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
239 *requirements* are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
240 restart on a worker blip), **progress states** ? which rule out fire-and-forget
Celery/RQ
241 for the long jobs. The *choice* should be made by prototyping in the
242 [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
243 - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
244 Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
245 async slice, with the lowest operational complexity and the easiest debugging.
246 - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
247 retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
248 because it adds **no new datastore** once we're on Postgres, but it *is* a new
249 dependency + operational/debugging surface; adopt only when long-job resumability/
250 observability demands it.
251 - **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
252 reserve for genuinely complex multi-stage pipelines at scale.
253
254 Record the decision criteria in #16: operational complexity, maturity, **debugging
255 experience for long GPU jobs**, and whether durability is worth a new dependency.
256 **Bias: start with the simplest thing that gives retries + status; graduate only on
257 evidence.**
258
259 ## 5A. Owned generative model (the video-QA assistant) & the data flywheel
260
261 The north-star capability: an **owned, self-hostable generative video model** ?
262 fine-tuned on sports video (including **consented, user-contributed** footage) ? that
263 **answers questions about rallies, shots, and technique**. This is the conversational
264 brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
265 is introduced **behind the same abstraction, with the paid API as an always-ready
266 fallback**.
267
268 > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
269 > assets are ***(a) the consented, badminton-specific video?QA dataset** and ***(b) the
270 > eval harness**. Build and message *those* ? not "we own an LLM."
271
272 This conversational surface (NL queries + coach-mode recommendations) must be
**localized
273 from the start** for the Bangalore beachhead ? the query *and its answer* in the user's
274 language. The locale foundation and the templated-answer approach are specified in

```

```

275     [`I18N_PLAN.md`](I18N_PLAN.md) (phase P4 ties to this section).
276
277     > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
in
278     > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
279     > serving / fallback* only ? never as a *training* data source.** Harvesting their
outputs
280     > to train our owned model would breach their "no competing model" terms (a
terms/copyright
281     > violation), so it is prohibited in both engineering and legal process. **Training
teachers
282     > are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
283     > distillation constraint below.
284
285     ### Core principle: AI is a pluggable provider with a paid fallback
286     The codebase already abstracts the cloud AI behind `provider_registry` /
287     `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
principle**:
288     - Every generative capability routes through the **provider registry**.
289     - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
290     or as the quality ceiling/teacher reference).
291     - This lets us **ship today on the known paid stack** and **swap in the owned model when
its quality clears a measured bar** ? with zero architectural churn and the paid path
292     always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
293     owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
294
295     ### The model / training layer (new sub-layer, fed by the scaffolding)
296     ```
297
298     Data curation          Training          Eval          Serving
299     (consented user vid   (fine-tune an open,   (extend the existing (self-host inference
300     + golden set +        commercially-usable   eval harness to QA   as a provider /
301     rally/segment/event  base VLM via          quality + the        ComputeBackend;
302     annotations ?        LoRA/QLoRA)          regression gate)     paid fallback)
303     video-QA pairs)
304     ```
305
306     This layer **reuses every other layer**: **Storage** holds the training corpus;
307     **Data** supplies labels (the `annotation` registry + `candidate_segments` +
308     golden set are already the labeling seam); **Compute/Orchestration** runs training and
309     inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
310     **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
311     which is why scaffolding comes first (see S6).
312
313
314     ### Base model & training (Apache-2.0 only ? licensing is decisive)
315     **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
316     (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
317     **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
318     Google-stack alternative to evaluate.
319     - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
320     is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
321     - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
322     permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
to
323     Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
324     model name + "Built with Llama" badge and a 700M-MAU ceiling. **gpt-oss** ?
Apache-2.0
325     but **text-only**, so usable only as an optional text-reasoning layer, never the video
326     encoder.
327
328     - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
329     avoids catastrophic forgetting of general video skills. Build the dataset by
330     converting
331     the existing rally/shot/segment annotations + golden set into templated **clip?QA

```

```

332     pairs**, expand synthetically with **open-model teachers only** (see constraint
below),
333     and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
?
334     **5k?20k** curated is a realistic band for this narrow domain.
335 - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
336     comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
have
337     sustained video-QA volume **+++ a defensible consented dataset +++** real per-call API
338     cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
339     is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
340
341     Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
```

342

```

343     ### Training backend ? managed service vs self-managed (the fine-tune "forge")
344     *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
```

345 smallest servable model, RL support, no-train DPA, and infra burden**.

346

```

347     **Decisive finding (researched 2026-06):** managed fine-tuning services are
348     **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
349     as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
350     video (+audio) training.** So for true video-QA, a self-managed framework is the
351     strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
352     serve) and keep full ownership. Managed services trade some of that for zero infra.
```

353

Option	Type	Video / VLM support	Smallest VL	RL	Export weights	Best for
ms-swift (ModelScope)	Self-managed (rented/own GPU)	Native video+audio	Qwen3-VL/Omni, 300+ MLLMs	any incl 7B	? DPO/GRPO	? full ownership
LLaMA-Factory	Self-managed	Qwen3-VL + 100+ VLMs; video datasets	any incl 7B	? ? ?	popular, well-documented	
Unsloth	Self-managed (1-GPU optimized)	Qwen3-VL LoRA (vision+lang; frames)	any incl 7B	? GRPO	? ?	cheapest first runs
Tinker (Thinking Machines)	Managed API (low-level)	Qwen3-VL 30B-A3B+ only	image input (video **unverified**)	30B-A3B MoE	? PPO/GRPO, custom loss	? adapters + ckpt; no-train ToS
Together AI	Managed (tune + serve)	Qwen3-VL / Gemma-3 / Llama-4; image data	verify partial	? adapters downloadable	managed Qwen3-VL + serving	
Fireworks AI	Managed (tune + serve)	Full Qwen2.5-VL (3B?72B SFT; LoRA@32B + LoRA-upload)	7B	partial	?? verify export (lock-in risk)	managed *small*-VL + serving
Predibase (Rubrik)	Managed	+++ VPC self-host	VLM instruct-tune	**LoRAX** multi-adapter	verify	? (VPC) ? **RFT/GRPO** flagship RL "creativity" + multi-adapter + privacy
OpenAI GPT-4o / Gemini tuning	~	Closed	? ? ? ?	? ?	no export	**non-fit ? violates "own the model"*

364

```

365     *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
366     control but are only worth it if you're already on that stack.)*
367
368     **Recommendation:**
369     - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
370     Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
371     video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
372     **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
373     the video requirement *and* serving economics.
374     - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
375     or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
376     adapter export.
```

```

377 - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
378 (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
379 level RL primitives).
380 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
381 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
382 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
383 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
384
385 **Verify before committing (any backend):** (1) **native video / frame-sequence**
386 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering
387 consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the
size
388 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
389
390 ### Training harness ? make it agent-runable (nanochat-style operability)
391 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
392 and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is
the
393 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
394 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
395 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
396 five properties are exactly what make training agent-runable, and we copy them:
397 ***(1)** one command, end-to-end; ***(2)** one complexity dial (sane auto-derived
defaults);
398 ***(3)** an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
399 ***(4)** reproducible + deterministic + cheap; ***(5)** minimal/readable.
400
401 **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
402 **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
403 `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
404 ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
405
406 **The harness = thin glue over components we already have** ? one declarative,
407 agent-invokable job (call it `train_speedrun`):
408 1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
409 `candidate_segments` + golden set ? the labeling seam in `backend/`).
410 2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an
orchestration
411 job (§5 / DEFERRED #16): one config, idempotent, logged.
412 3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
413 **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
414 4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
415 if it beats the bar**.
416
417 So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
418 provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
419 wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
420
421 > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
422 > set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
423 > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
424 > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
425
426 ### ?? Teacher / distillation constraint (the single biggest legal landmine)
427 You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
428 video-QA assistant is squarely a "competing model": **Gemini** ("may not use the

```

```

429 Services to develop models that compete"), **OpenAI** ("may not use Output to develop
430 models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
431 safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL /
432 InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime
433 fallback** ? serving answers is fine, *harvesting its outputs to train the model that
434 replaces it is not*. Write a hard ***no training on competitor-API outputs*** rule into
435 eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
436 the teacher is *open models*, not the paid APIs.)
437
438 ### Data flywheel, personalization & consent tiers
439 Two complementary loops, each with its **own** opt-in (training consent is separate from
440 processing consent ? §7):
441
442 - **Global model (shared ? raises the floor).** Train the base owned model on an
443 **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
bucket
444 (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
445 better shared model ? better highlights/answers ? more reason to contribute.* Higher
446 consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.
447
448 - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
exploration.**
449 A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
450 consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
451 Together / Predibase). Why it's compelling:
452 - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
453 styles ? materially better detection/highlights for them.
454 - **Cleanest consent story:** ***improve **your** detection using **your** data*** is
far
455 lower-friction and lower-risk than contributing to a global model ? the data
improves
456 **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
the
457 pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
video,
458 your storage, your compute ? and now **your model**.* With BYO-compute the adapter
can be
459 trained/served on the user's own GPU and **never leave their boundary**; **deletion
=
460 drop the adapter** (clean erasure).
461 - **Open questions (still a hunch):** cold-start (need enough of the user's labeled
data
462 before a personal adapter beats the base ? fall back to the shared model meanwhile);
a
463 **per-user held-out eval** to prove the adapter helps (don't ship a regression);
base +
464 optional global-adapter + per-user-adapter stacking; per-user training cost at
scale.
465 Worth a spike once the harness + golden eval exist.
466
467 **Two consent tiers, then:** (a) "improve **my own** detection with **my** data"
468 (per-user, low friction) and (b) "contribute to the **shared** model" (global,
higher
469 bar). **Default for both is off.**
470
471 ## 6. Phased roadmap
472
473 | Phase | Deliverable | Build / reuse |
474 |---|---|---|
475 | **P0** | ? **DONE ? Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
`user_version` ladder; single-worker `ThreadPoolExecutor`; `/api/process` ? 202 + job_id; `GET
/api/jobs/{id}` status |
476 | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +
history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |

```

```

477 | **P2** | StorageBackend registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
478 | **P3** | ComputeBackend registry + BYO-compute** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
479 | **P4** | Rich rally/shot semantic metadata & tagging (the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
outcome (winner / forced / unforced error), context (e.g. open court but error) ?
query-API filters + HighlightMetadata + a human correction/review loop feeding the golden
set via the annotation registry + candidate_segments |
480 | **P5** | Owned generative video-QA model ($5A) | global flywheel per-user
adapters; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
owned_vlm provider with Gemini paid fallback; extend the eval harness to QA quality |
481 | **P6** (visionary) | Gait / biomechanics under a supervising coach |
deliberately back-burner; technique feedback; highest Art. 9 bar ? gate behind explicit
biometric + parental consent ($7) |
482
483 P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
484 deliver the "plug-and-play storage/compute" ask. P4 (rich semantic metadata) is the
485 immediate next value stage once rally detection + highlights are solid** ? e.g.
"rallies
486 that ended in a smash or backhand drive," "rallies where the court was open but the
487 player made an unforced error." P5 owns the model; P6 (gait) is the visionary
488 horizon, intentionally deferred.**
489 Sequencing principle (owner directive): make the scaffolding (P0?P4) rock-solid
490 first, keep the known paid stack as an always-ready fallback**, and only then shift
491 focus to owned-model quality (P5). The provider registry keeps the paid path one
492 config flip away.
493
494 ## 7. Privacy & legal (gait raises the bar ? design for it now)
495
496 - The load-bearing nuance: biometric data is special-category only when used to
497 uniquely identify a person** (GDPR Art. 9). Highlight/shot detection on players a
498 coach already knows is general analytics ? generally not Art. 9.** But
499 gait/biomechanics that builds a per-athlete identifying template tips into Art. 9**
500 and needs explicit, granular consent referencing biometric data. Engineer gait
501 with that boundary (consent gate, purpose limitation, no cross-account
identification);
502 decide whether it runs locally/on-tenant only (strong posture) vs our cloud. (EDPB
503 Guidelines 3/2019 on video devices apply.)
504 - Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-
order,
505 not an afterthought. Model consent explicitly per relationship (user?subject,
coach?athlete
506 where the overlay applies, parent?minor).
507 - BYO-storage is the central privacy/liability lever:** if raw video stays in the
508 tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
509 what we hold ? shrinking breach surface, residency, and erasure obligations.
510 - Paid-LLM no-train (C4): use Gemini via Vertex AI (contractual no-training),
511 not the consumer/free tier; set short/zero retention, document it in the DPA, and
512 surface "your footage is never used to train AI" as a trust feature for academies. For
513 the gait phase, prefer keeping biometric inference off third-party APIs entirely.
514 - Training consent is a distinct, higher bar (for the owned model, $5A).** Training on
515 user video needs a separate, explicit, opt-in training licence ? not bundled with
516 processing consent ? and it compounds the biometric (Art. 9) and minor-consent
517 issues above. Train only on explicitly contributed/licensed video; never train on
518 what merely passes through a tenant's bucket. Exclude minors' video from the
training
519 pool by default**, keep the training corpus a separate, firewalled, opt-in sub-
pool**,
520 and log per-clip consent/provenance for revocation + audit. Incentives must not
make
521 consent non-free (the product stays fully usable without contributing). And the
522 teacher-distillation rule ($5A): never train on paid-LLM outputs. Two consent
tiers

```

523 (\$5A):** *(a)* "improve **my own** detection with **my** data" ? a per-user adapter
 trained
 524 only on the user's own video (lowest friction; no cross-user mixing; deletion = drop
 the
 525 adapter); *(b)* "contribute to the **shared** model" ? pooled corpus, higher bar.
 Default
 526 for both is **off**.
 527 - **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across
 DB +
 528 storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends
 here).
 529
 530 ## 8. Key decisions for the owner (the forks)
 531
 532 1. **Hosted-first vs BYO-first.** Recommended: **hosted control-plane first** (P0?P1)
 with
 533 **BYO-storage soon after** (P2); **BYO-compute later** (P3) as a power-
 user/enterprise
 534 option ? it's the highest-complexity, highest-security item.
 535 2. **Database now or later.** Move to **Postgres at P1** for the hosted path, or stay
 536 SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
 537 regardless.)
 538 3. **Auth: build vs buy** (Auth0 / Clerk / Supabase Auth / Cognito).
 539 4. **Orchestration tech** (see research recommendation) ? bias to the simplest that
 gives
 540 retries + status.
 541 5. **Gait processing location** ? local/on-tenant (privacy-max) vs cloud (capability-
 max).
 542 6. **How far to push BYO-compute** given its security/ops complexity.
 543 7. **Owned-model base + licensing (\$5A).** Which commercial-OK video VLM to build on
 544 (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial
 545 derivative is permitted ? see COMMERCIALIZATION C14.
 546 8. **Training-data sourcing & consent.** Opt-in contribution terms, incentives, and the
 547 teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
 548 9. **When to start the owned model** ? strictly after the scaffolding + flywheel are
 549 solid (the explicit directive), with the paid provider as the standing fallback.
 550
 551 **Risks:** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +
 552 tests); BYO-compute security; **biometric/minor liability** (get this right before any
 553 gait feature ships); **training on user video without proper opt-in/licence**, and
 554 **base-model / teacher-distillation licence violations** ; cost of hosted GPU (training
 555 *and* inference); **premature model-building** before data/eval are ready (mitigated by
 556 keeping the paid fallback and gating P5 behind the flywheel).
 557
 558 ## 9. Sources
 559
 560 External best-practices research (2026-06-07) backing the recommendations above. Each
 561 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and
 562 should be re-validated before committing budget.
 563
 564 **Storage abstraction:** [fsspec](https://fsspec.com/) .
 [cloudpathlib](https://github.com/drivendataorg/cloudpathlib) .
 [smart_open](https://github.com/piskvorky/smart_open) . [Apache
 Libcloud](https://libcloud.apache.org/) . [gdrive-fsspec](https://github.com/fsspec/gdrive-
 fsspec) . [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) . [AWS S3 presigned
 URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) . [token-
 based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) .
 [signed URLs across clouds](https://medium.com/@shaikhsharwan49/signed-urls-in-aws-gcp-and-azure-
 a-beginner-friendly-guide-e2e549425865)
 565
 566 **Multi-tenancy:** [Bytebase: multi-tenant DB
 patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/) .
 [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-
 architecture) . [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-
 tenancy-in-postgres) . [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-

tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

567

568 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) · [Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) · [GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank: Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-deployment) · [DevPro: Python task queues 2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) · [Hatchet](https://github.com/hatchet-dev/hatchet)

569

570 ****Privacy / legal:**** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

571

572 ****Comparable products:**** [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

573

574 ****Training backends & harness (§5A):**** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

575

576 ****Owned model & licensing (§5A):**** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI

terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlmtrl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

577

578 ## See also

579 - COMMERCIALIZATION.md ? C12 (SaaS foundations), C4 (paid-LLM), economics.

580 - DEFERRED_HARDENING_PLAN.md ? #16 async job model (the P0 infra slice), #7 packaging.

581 - PMF_MARKET_RESEARCH.md ? academy/coach beachhead (who this serves).

582 - CODE_MAP.md ? current architecture + the registry extension points.

583 - ROADMAP.md ? offline self-improving rally detection (the pipeline's own trajectory).

584

11. assistant (2026-06-20T08:59:42.802Z)

Now let me read SETUP_NEW_MACHINE.md and the deploy scripts to get the actual droplet/machine layout.

12. user (2026-06-20T08:59:43.758Z)

```
1      # Setting up a new machine ? zero ? training & inference
2
3      **The canonical "spin up a box and make the models ready" runbook.** Copy-pasteable;
followable by
4      any human. DigitalOcean is the worked example (GCP/AWS/Paperspace are analogous ? only
the provision
5      step differs).
6
7      > **Last verified:** 2026-06-20 on a DigitalOcean Droplet (Ubuntu 24.04, 2 vCPU / 3.8
GB, **no GPU**).
8      > Bootstrap succeeded (torch 2.12.1+cpu); full mocked suite **922 passed / 11 skipped**;
```

a **secret-free**

```
9      > CPU inference (no GPU, no API key) detected **2 rallies** and persisted them to
SQLite.
10     >
11     > ? Assumes the DECOUPLED_COMPUTE PRs are merged: the CPU-mock detector seam,
`deploy/digitalocean/bootstrap.sh`,
12     > the storage layer, and the worker dispatcher (#251). The **bucket** steps ($7) are
config-driven and
13     > **verified against real DO Spaces + GCS** ? see the appendix.
14
15     ## What you'll have at the end
16
17     A machine that can: run the test suite (green), run **secret-free CPU inference**
(rallies, no GPU/keys),
18     run **real Gemini inference** (with a key), and run **Gen-0 model training** ? plus
pointers to the GPU /
19     native-detector path and bucket-driven train/infer.
20
21     ---
22
23     ## 1. Credentials to create once (least-privilege)
24
25     | # | Credential | Where | Needed for |
```

```

26 |---|---|---|---|
27 | 1 | **SSH keypair** (public key added to the provider) | `ssh-keygen -t ed25519 -f
~/.ssh/rally` ? paste `.pub` into DO ? Settings ? Security | logging into the box |
28 | 2 | **DigitalOcean API token** (read+write) | DO ? API ? Tokens | `doctl` provisioning
(optional ? you can click-create) |
29 | 3 | **GitHub repo read access** | a read-only **deploy key** *or* a fine-grained
**PAT** (`Contents: Read`) | `git clone` on the box (or use the `scp` path in $3) |
30 | 4 | **GEMINI_API_KEY** | Google AI Studio / Vertex | **real** Gemini inference only
($6b). The secret-free path ($6a) needs none. **Rotate before sharing.** |
31 | 5 | **Bucket creds** ? DO Spaces key/secret, or GCS SA JSON, or Drive SA JSON + folder
| provider console | bucket-driven train/infer ($7, *pending PR-B*) |
32 | 6 | **GPU box** (Paperspace free M4000 / GPU Droplet) | provider console | real
WASB/TrackNet detectors + faster runs ($8) |
33
34 **Secret hygiene:** never put a secret in the repo, a PR, or `config.json`. Set them as
env vars on the box
35 (or a git-ignored `.env`). Scope each as narrowly as the UI allows and rotate/revoke
when done.
36
37 ---
38
39 ## 2. Provision the machine
40
41 - **CPU box** (suite + secret-free stub inference + Gen-0 training plumbing): cheapest
**Ubuntu 24.04**
42 Droplet, **2 vCPU / 4 GB**, region nearest you, **attach the SSH key** (#1). ~$0.06/hr
? destroy when done.
43 - **GPU box** (real WASB/TrackNet detectors ? real trajectory extraction + training): a
**DO GPU Droplet**.
44 ? You **cannot add a GPU to a CPU Droplet** ? it's a separate machine type, in **GPU
regions**
45 (`nyc2`/`tor1`/`atl1`, **not** `blr1`), with **dual local NVMe**. **RTX 4000 Ada (20
GB, ~$0.76/hr) **is
46 plenty for WASB + Gen-0; the `gpu-*-base` image preinstalls the NVIDIA driver. **Co-
locate the Space in the
47 GPU box's region **for cheap corpus-pull.
48
49 CLI provision (after `doctl auth init` with token #2 ? **verify size/image slugs first**
via `doctl compute size list`):
50 ```bash
51 # CPU box:
52 doctl compute droplet create rally-box --image ubuntu-24-04-x64 --size s-2vcpu-4gb \
53 --region blr1 --ssh-keys <your-key-fp> --wait
54 # GPU box (RTX 4000 Ada):
55 doctl compute droplet create rally-gpu --image gpu-h100x1-base --size gpu-4000adax1-20gb
\
56 --region tor1 --ssh-keys <your-key-fp> --wait
57 doctl compute droplet list # note the IP -> ssh -i ~/.ssh/rally root@<IP>
58 ```
59
60 ---
61
62 ## 3. Get the code onto the box
63
64 **Option A ? git clone** (private repo, so authenticate with #3):
65 ```bash
66 ssh -i ~/.ssh/rally root@<IP>
67 git clone git@github.com:avidullu/badminton-highlight-indexer.git ~/rally # via deploy
key
68 # or: git clone https://<PAT>@github.com/avidullu/badminton-highlight-indexer.git
~/rally
69 cd ~/rally
70 ```
71
72 **Option B ? push a local checkout** (no GitHub access needed; what the maintainer

```

```

used):
73     ```bash
74     # on your machine, from the repo root:
75     git archive --format=tar.gz -o /tmp/rally.tgz HEAD
76     scp -i ~/.ssh/rally /tmp/rally.tgz root@<IP>:/root/
77     ssh -i ~/.ssh/rally root@<IP> 'mkdir -p ~/rally && tar -xzf ~/rally.tgz -C ~/rally'
78     ```
79
80     ---
81
82     ## 4. Bootstrap (one command)
83
84     ```bash
85     cd ~/rally
86     ./deploy/digitalocean/bootstrap.sh           # CPU box
87     # or
88     ./deploy/digitalocean/bootstrap.sh --gpu     # GPU box (installs CUDA torch)
89     ```
90
91     It mirrors CI: installs `ffmpeg` + opencv runtime, creates `.venv`, installs CPU-or-CUDA
92     `torch`, then
93     `pip install -e .[dev]`, and prints a device check (`cuda_available True/False`).
94
95     <details><summary>Manual equivalent (if you can't run the script)</summary>
96
97     ```bash
98     sudo apt-get update
99     sudo apt-get install -y python3 python3-venv python3-pip git ffmpeg libgl1
100    sudo apt-get install -y libgl1 2.0-0 || sudo apt-get install -y libgl1 2.0-0
101    python3 -m venv .venv && . .venv/bin/activate && python -m pip install -U pip
102    pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu # or
103    .../whl/cu124 on GPU
104    pip install -e .[dev]
105    ```
106
107     </details>
108
109     ---
110
111     ## 5. Verify the machine (the smoke gate)
112
113     ```bash
114     . .venv/bin/activate
115     python -m pytest -q           # expect all green (922 passed / 11 skipped on the current
116     tree)
117     ```
118
119     A green suite proves the whole pipeline runs on this box with **no GPU, no WSL, no
120     bucket, no API key**.
121
122     ---
123
124     ## 6. Run inference
125
126     ### 6a. Secret-free CPU inference (no GPU, no API key) ? the "mock a GPU with a CPU"
127     path
128
129     Drop a permissive smoke `config.json` in the repo root (the CLI reads `./config.json`
130     from the cwd):
131
132     ```bash
133     cd ~/rally && cat > config.json <<'JSON'
134     {
135         "sport": "badminton",
136         "indexing": { "default_segmenter": "wasb_hybrid", "detector_impl": "stub",
137         "skip_ai_handoff": true },
138         "validation": { "min_fps": 0, "min_blur_threshold": 0, "max_scene_cuts_per_minute":
139         1000,

```

```

129             "relevance": { "court_pixels_min_ratio": 0.0 } }
130     }
131     JSON
132
133     # a throwaway 12s 1280x720 test clip
134     ffmpeg -y -f lavfi -i testsrc=duration=12:size=1280x720:rate=30 -pix_fmt yuv420p
/root/sample.mp4
135
136     # run ? env -u drops any Gemini key to prove none is needed
137     env -u GEMINI_API_KEY RALLY_DETECTOR_IMPL=stub \
138         indexer --video-path /root/sample.mp4 --segmenter wasb_hybrid --output-dir ~/rally/out
139     ```
140     Expect: `detector_impl=stub ? CPU mock runner`, `Phase 3 SKIPPED (fully-local)`, and
141     `Indexed 2 play segments` in `~/rally/out/sports_indexer.db`. *(`detector_impl=stub`
swaps the GPU+WSL
142     detector for a deterministic CPU one; `skip_ai_handoff=true` makes candidate windows the
rallies, so no
143     Gemini handoff happens.)*
144
145     ### 6b. Real inference (Gemini segmenter)
146
147     ```bash
148     export GEMINI_API_KEY=""           # secret stays in the env, never in
the repo
149     # use a real badminton clip (must pass validation: ?1280x720, ?30fps, a green court,
etc.)
150     indexer --video-path /root/match.mp4 --segmenter gemini --output-dir ~/rally/out
151     # or the hybrid: --segmenter wasb_hybrid (WASB needs a GPU box, S8) with
skip_ai_handoff=false
152     ```
153
154     ---
155
156     ## 7. Object storage ? bucket-driven train/infer (S3-compatible / GCS / Drive)
157
158     The worker (`python -m backend.worker`) reads/writes by **URI**, so you choose a backend
with **config +
159     credentials ? no code change**. A bare path / `local://` is a passthrough (today's
behaviour); `s3://` and
160     `gcs://` go to a bucket. Install only the extra you use (cloud SDKs are lazy-imported).
161
162     > **Verified end-to-end 2026-06-20:** the worker bucket round-trip (pull video ? infer ?
push outputs) passed
163     > against **real DigitalOcean Spaces** (sgp1), a real **GCS** client (emulator), and
**MinIO**.
164
165     ### 7a. S3-compatible ? AWS S3 / DigitalOcean Spaces / Cloudflare R2 / MinIO
166
167     One backend for all four ? they differ only by `endpoint_url` + region. `pip install -e
.[storage-s3]`.
168
169     **Credentials** ? `~/aws/credentials` (boto3's default chain; **never** in the repo):
170     ```ini
171     [default]
172     aws_access_key_id = <key>
173     aws_secret_access_key = <secret>
174     ```
175     **Config** ? `config.json` `storage.s3`:
176     ```jsonc
177     "storage": { "backend": "local", "s3": {
178         "endpoint_url": "https://sgp1.digitaloceanspaces.com", // DO Spaces (VERIFIED);
region = the Space's, e.g. sgp1
179         "region": "sgp1"
180         // AWS S3 : omit endpoint_url; set region (e.g. ap-south-1)
181         // R2      : endpoint_url https://<account>.r2.cloudflarestorage.com ; region "auto"

```

```

182      // MinIO : endpoint_url http://host:9000 ; region us-east-1 ; "addressing_style":
"path"
183    }}
184    ```
185    **Run** (secret-free flavor; use `--segmenter gemini` for real detection):
186    ```bash
187    env -u GEMINI_API_KEY RALLY_DETECTOR_IMPL=stub python -m backend.worker infer \
188      --video-uri s3://<bucket>/videos/clip.mp4 --out-uri s3://<bucket> \
189      --segmenter wasb_hybrid --set indexing.skip_ai_handoff=true
190    ```
191
192    ### 7b. Google Cloud Storage
193
194    `pip install -e .[storage-gcs]`. **Credentials** ? a service-account JSON (SA with
*Storage Object Admin*):
195    ```bash
196    export GOOGLE_APPLICATION_CREDENTIALS=/path/to/sa.json
197    ```
198    **Config** ? `storage.gcs.project` (no endpoint ? default `googleapis.com`):
199    ```jsonc
200    "storage": { "backend": "local", "gcs": { "project": "<gcp-project-id>" } }
201    ```
202    Same `worker infer` command with `gcs://<bucket>/` URIs. *(On-box test with no GCP
creds:
203    `pip install gcp-storage-emulator && gcp-storage-emulator start --port 9023 --in-memory`
+
204    `export STORAGE_EMULATOR_HOST=http://127.0.0.1:9023`.)*)
205
206    ### 7c. Google Drive *(corpus source ? follow-up backend)*
207
208    `gdrive://` is recognised; the backend is a focused follow-up (for the read-only golden
corpus).
209
210    **Bucket layout:** `videos/ labels/ manifest.json weights/ outputs/<video_id>/` (see
`docs/DECOUPLED_COMPUTE/04`).
211
212    ---
213
214    ## 8. Run training (Gen-0 rally head)
215
216    CPU-runnable, minutes ? but it needs a **labeled corpus** (videos + `.rallies.csv`
labels + a manifest),
217    which a fresh box doesn't have. Bring it via $7's bucket (`worker train`, pending) or
`scp` it up, then:
218    ```bash
219    python -m training.gen0.harness --manifest <manifest.json> --floor <baseline> --version
0.1.0
220    ```
221    The harness runs train ? eval ? ship-gate and emits a versioned weight bundle. *(The
Gen-2 VLM is a
222    separate, GPU-heavy, separately-budgeted effort ? out of scope here.)*
223
224    ---
225
226    ## 9. GPU box ? port procedure (NVMe + native WASB)
227
228    The GPU box is **cattle, not a pet**: durable state lives in the bucket + git, so a
fresh box is running in
229    minutes. Port = provision ($2) ? code on ($3) ? NVMe ? `bootstrap.sh --gpu` ?
`gpu_setup.sh` ? pull corpus.
230
231    ```bash
232    # on the box, after $3 put the code at ~/rally:
233    cd ~/rally
234    sudo bash deploy/digitalocean/mount_scratch.sh && export TMPDIR=/scratch #

```

```

detect+mount the spare NVMe
235     ./deploy/digitalocean/bootstrap.sh --gpu                                # CUDA
torch; expects cuda_available True
236     . .venv/bin/activate && pip install -e .[storage-s3]
237     bash deploy/digitalocean/gpu_setup.sh                                # native
WASB conda env (no WSL) ? M3.5 enabler
238     # secrets (the only two non-reproducible items): Gemini key + bucket creds
239     mkdir -p /root/.keys && printf '%s' '<rotated-key>' > /root/.keys/GEMINI_API_KEY &&
chmod 600 /root/.keys/GEMINI_API_KEY
240     install -m600 /dev/stdin ~/.aws/credentials <<'INI'
241     [default]
242     aws_access_key_id = <key>
243     aws_secret_access_key = <secret>
244     INI
245     ```
246
247     - **NVMe scratch** (`mount_scratch.sh`): DO GPU Droplets ship a second raw NVMe; the
script formats+mounts it
248         at `/scratch` and the pipeline's sync-first decode runs there (`export
TMPDIR=/scratch` or `--scratch /scratch`).
249         Ephemeral, not in fstab. On a CPU box it just makes `/scratch` on the boot disk.
250     - **Native WASB** (`gpu_setup.sh`): replicates `docs/TRACKNET_WSL_SETUP.md`'s conda env
**natively on Linux**
251         (no WSL) ? the M3.5 enabler. ? Needs `wasb_badminton_best.pth.tar` present and the
**WASB-C3** weight
252         provenance cleared before sharing any derived model.
253     - **The detector itself:** real `--segmenter wasb_hybrid` extraction needs the Linux-
native `DetectorRunner`
254         (tracked M3.5, `docs/DECOUPLED_COMPUTE/04`). Until it lands, validate the train/infer
plumbing with
255         `detector_impl=stub`.
256     - **Teardown** = `doctl compute droplet delete rally-gpu`. Re-spin tomorrow identically.
257
258     ---
259
260     ## 10. Teardown
261
262     ```bash
263     rm -f ~/rally/config.json ~/sample.mp4                # remove the throwaway smoke artifacts
264     doctl compute droplet delete rally-box                # or destroy in the console
265     ```
266     Revoke any temporary token/key (#2?#5) and delete the SSH key from the provider when the
box is gone.
267
268     ---
269
270     ## Appendix ? verification log
271
272     | Step | Command | Result (2026-06-20, DO Ubuntu 24.04, 2 vCPU, no GPU) |
273     |---|---|---|
274     | Bootstrap | `bootstrap.sh` | ? torch 2.12.1+cpu, deps + editable build |
275     | Smoke | `pytest -q` | ? 922 passed, 11 skipped (49s) |
276     | Secret-free infer (local) | $6a | ? 2 rallies, persisted to SQLite, `GEMINI_API_KEY`
unset |
277     | **S3 worker round-trip ? real DO Spaces (sgp1)** | $7a | ? pulled
`s3://rally-e2e/videos/?` ? 2 rallies ? pushed `outputs/<md5>/intervals.csv` |
278     | **GCS worker round-trip** (emulator) | $7b | ? pulled `gcs:///?` ? 2 rallies ? pushed
`gcs:///?/outputs/<md5>/?` |
279     | S3 backend vs MinIO (put/get/list/stat + presigned GET) | $7a | ? |
280

```

13. user (2026-06-20T08:59:43.930Z)

```

1     # Testing the rally pipeline on DigitalOcean
2

```

```

3      A storage-free, secret-free way to get the pipeline running on a DigitalOcean (or
Paperspace)
4      machine ? the compute half of
[docs/DECOUPLED_COMPUTE/](../docs/DECOUPLED_COMPUTE/README.md).
5      The object-store storage shim is intentionally out of scope here** (owner is
finalizing it); this
6      runbook only proves compute works on the box.
7
8      > ? The commands below mirror .github/workflows/ci.yml (Ubuntu) so the box matches CI,
but the
9      > bootstrap.sh script has not yet been run on a real DO box ? treat it as a
tested-by-mirroring
10     > starting point and adjust as you go.
11
12     ## 1. Credentials to create (least-privilege)
13
14     | # | Credential | Where to create | Scope | Needed |
15     |---|---|---|---|---|
16     | 1 | SSH keypair | local ssh-keygen; add the public key to DO (Settings ?
Security) / Paperspace | machine login | now |
17     | 2 | DigitalOcean API token | DO console ? API ? Tokens ? Generate (Personal Access
Token) | read + write (Droplets/Spaces); scope down if the UI allows | now (for doctl;
optional if you click-create the box) |
18     | 3 | Paperspace / Gradient API key | console.paperspace.com ? API Keys | account |
for the GPU box (free M4000 / cheap A4000 live under Paperspace) |
19     | 4 | GitHub repo read access on the box | a read-only deploy key (repo ?
Settings ? Deploy keys) or a fine-grained PAT with Contents: Read | this repo only |
now (to git clone) |
20     | 5 | DO Spaces keys (access key + secret) | DO console ? API ? Spaces Keys |
one bucket/prefix | defer ? only when the storage shim + a real data test land |
21     | 6 | GEMINI_API_KEY | Google AI Studio (or Vertex) | the Gemini segmenter |
later ? only to test the Gemini path; the stub/WASB path needs none. Rotate the old key
that was shared in chat. |
22
23     The "API key that drives the GPU" in the North-Star sense is #2 (DO token) +
#3 (Paperspace key) ?
24     they let you (or doctl/a control plane) spin up and drive GPU machines. The per-
invocation serverless
25     endpoint key (RunPod / Cloud Run) is a later M3 concern.
26
27     Secret hygiene: never bake any of these into the repo or a Docker image. Put them in
the box's
28     environment / a .env that is git-ignored, or the provider's secret store. Prefer
scoping #2 and #4 to the
29     narrowest resource the UI allows.
30
31     ## 2. Bring up a box
32
33     - CPU smoke (M0/M1-cpu): the cheapest Droplet (a basic shared-CPU Ubuntu
22.04/24.04) ? or, free, the
34     Paperspace Gradient M4000 GPU machine (unbilled on the Free plan).
35     - GPU test (M1 / M3.5 WSL?native parity): a Paperspace GPU (free M4000, then
P4000/RTX4000/A4000
36     on credit) or a GPU Droplet. All are Ubuntu + NVIDIA driver + CUDA.
37
38     doctl (optional) once #2 is set: doctl auth init then doctl compute droplet create
? Or just create it
39     in the console and copy the IP.
40
41     ## 3. Bootstrap + smoke (on the box)
42
43     ``bash
44     ssh user@<box-ip>
45     git clone <repo> # via deploy key (#4)
46     cd badminton-highlight-indexer

```

```

47
48     ./deploy/digitalocean/bootstrap.sh           # CPU box
49     # or
50     ./deploy/digitalocean/bootstrap.sh --gpu      # GPU box (installs CUDA torch)
51
52     . .venv/bin/activate && python -m pytest -q    # storage-free, secret-free smoke (~900
passed)
53     ...
54
55     **Acceptance:**
56     - CPU box ? the mocked suite passes (the whole pipeline runs with no
GPU/WSL/bucket/key).
57     - GPU box ? the bootstrap device check prints `cuda_available True` + the GPU name (M1
acceptance).
58
59     Once [#247](https://github.com/avidullu/badminton-highlight-indexer/pull/247) (the CPU-
mock detector seam)
60     is merged, you can also drive the trajectory segmenter GPU-free end-to-end:
61     `RALLY_DETECTOR_IMPL=stub python -m pytest tests/test_stub_runner.py -q`.
62
63     ## 4. Next compute steps (not in this runbook)
64
65     - **M3.5 WSL?native detector parity** ? replicate the `TRACKNET_WSL_SETUP.md` WASB conda
env *natively*
66     on the GPU box (no WSL), add a `LinuxNativeRunner`, and parity-test its trajectory CSV
vs the WSL box.
67     - **`train` / `infer` dispatcher + the bucket** ? gated on the storage shim (owner-
finalizing).
68

```

14. assistant (2026-06-20T08:59:48.183Z)

Now let me read the deploy scripts to get the actual ports, paths, and secrets handling.

15. user (2026-06-20T08:59:49.425Z)

```

1     #!/usr/bin/env bash
2     #
3     # Bootstrap a DigitalOcean / Paperspace Ubuntu box to run the rally pipeline.
4     #
5     #     ./deploy/digitalocean/bootstrap.sh           # CPU box (Droplet)           ? M0 /
M1-cpu
6     #     ./deploy/digitalocean/bootstrap.sh --gpu      # GPU box (Paperspace / GPU Droplet) ?
M1 / M3.5
7     #
8     # Mirrors .github/workflows/ci.yml (Ubuntu) so the box matches CI exactly.
9     # STORAGE-FREE + SECRET-FREE: proves the pipeline runs on the box via the mocked
10    # test suite. No bucket, no Gemini key, no WSL required. See
11    ../../docs/DECOUPLED_COMPUTE/.
12
13    set -euo pipefail
14
15    GPU=0
16    [ "${1:-}" = "--gpu" ] && GPU=1
17
18    echo ">> [1/4] System libraries (ffmpeg + opencv runtime)"
19    sudo apt-get update
20    sudo apt-get install -y python3 python3-venv python3-pip git ffmpeg libgl1
21    # opencv needs libgl2.0-0; the package name differs across Ubuntu releases (t64 on
22    24.04).
23    sudo apt-get install -y libgl2.0-0t64 || sudo apt-get install -y libgl2.0-0
24
25    echo ">> [2/4] Python venv (.venv)"
26    python3 -m venv .venv
27    # shellcheck disable=SC1091
28    . .venv/bin/activate

```



```

26     python -m pip install --upgrade pip
27
28     echo ">> [3/4] torch (${[ "$GPU" = 1 ] && echo CUDA || echo CPU}) + package (editable,
dev tooling)"
29     if [ "$GPU" = 1 ]; then
30         # CUDA 12.4 wheels. NOTE: verify your GPU arch is supported by this build.
31         # The Paperspace free M4000 is Maxwell (sm_52) ? old archs are sometimes dropped;
32         # if `torch.cuda.is_available()` is False on an M4000, use a P4000/RTX4000 or pin
33         # an older torch. (docs/DECOUPLED_COMPUTE/03 + 06.)
34         pip install torch torchvision --index-url https://download.pytorch.org/whl/cu124
35     else
36         pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
37     fi
38     # Runtime deps + pytest/ruff/mypy. The 'reporting' extra (private obsreport over SSH)
39     # is intentionally omitted ? it is a soft dep and its tests skip when absent.
40     pip install -e .[dev]
41
42     echo ">> [4/4] Device check"
43     python - <<'PY'
44     import torch
45     ok = torch.cuda.is_available()
46     print("torch", torch.__version__, "| cuda_available", ok,
47           "| device", (torch.cuda.get_device_name(0) if ok else "CPU only"))
48     PY
49
50     cat <<'EOF'
51
52     >> Bootstrap complete. Run the storage-free, secret-free smoke (mocked suite):
53
54         . .venv/bin/activate && python -m pytest -q
55
56         Expect ~900 passed. That proves the whole pipeline runs on this box with no GPU,
57         no WSL, no bucket and no API key. (On a GPU box the device check above should
58         print cuda_available True.)
59     EOF
60

```

16. user (2026-06-20T08:59:49.903Z)

```

1     #!/usr/bin/env bash
2     #
3     # Native (no-WSL) WASB shuttle-detector setup for a Linux GPU box ? the M3.5 enabler for
4     # real trajectory extraction. Replicates docs/TRACKNET_WSL_SETUP.md's WSL conda env
5     # *natively* on Linux. Run AFTER ./bootstrap.sh --gpu (which provides CUDA torch for the
6     # main venv); this builds a SEPARATE conda env for WASB (torch 1.11) ? never co-
installed
7     # with the indexer's torch (framework conflict).
8     #
9     # Requires an NVIDIA GPU + driver (the D0 `gpu-*-base` image ships it).
10    #
11    # ? WASB-C3: the pretrained badminton weights' license / source-dataset provenance is
NOT
12    # verified here. Fine for internal extraction + Gen-0 bootstrapping; clear it before
sharing
13    # any derived trajectories or model. See docs/DECOUPLED_COMPUTE/06.
14    set -euo pipefail
15
16    MODELS_DIR="${RALLY_MODELS_DIR:-$HOME/models}"
17    CONDA_DIR="${CONDA_DIR:-$HOME/miniconda3}"
18    WASB_REPO="$MODELS_DIR/WASB-SBDT"
19    mkdir -p "$MODELS_DIR"
20
21    echo "[gpu] [1/5] miniconda (native Linux)"
22    if [ ! -x "$CONDA_DIR/bin/conda" ]; then
23        curl -sSL https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -o

```

```

/tmp/mc.sh
24     bash /tmp/mc.sh -b -p "$CONDA_DIR" && rm -f /tmp/mc.sh
25     fi
26     # shellcheck disable=SC1091
27     source "$CONDA_DIR/etc/profile.d/conda.sh"
28
29     echo "[gpu] [2/5] WASB-SBDT repo"
30     if [ ! -d "$WASB_REPO/.git" ]; then
31         git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO"
32     fi
33
34     echo "[gpu] [3/5] conda env 'wasb' (py3.8 + torch 1.11.0+cu113 ? the verified WASB
stack)"
35     conda env list | grep -q '/envs/wasb$' || conda create -y -q -n wasb python=3.8
36     conda activate wasb
37     python -m pip install -q torch==1.11.0+cu113 torchvision==0.12.0+cu113 \
38         --extra-index-url https://download.pytorch.org/whl/cu113
39     [ -f "$WASB_REPO/requirements.txt" ] && python -m pip install -q -r
"$WASB_REPO/requirements.txt" || true
40
41     echo "[gpu] [4/5] pretrained badminton weights"
42     WEIGHTS="$WASB_REPO/pretrained_weights/wasb_badminton_best.pth.tar"
43     if [ -f "$WEIGHTS" ]; then
44         echo "[gpu]         found $WEIGHTS"
45     else
46         echo "[gpu]         !! MISSING $WEIGHTS ? fetch wasb_badminton_best.pth.tar
(Drive/bucket) before extraction."
47     fi
48
49     echo "[gpu] [5/5] GPU sanity check (in the wasb env)"
50     python - <<'PY'
51     import torch
52     ok = torch.cuda.is_available()
53     print("[gpu]         torch", torch.__version__, "| cuda_available", ok,
54         "| device", (torch.cuda.get_device_name(0) if ok else "CPU"))
55     assert ok, "CUDA not available in the wasb env ? check the GPU driver / base image"
56     PY
57
58     echo "[gpu] native WASB env ready.  Activate with:  source
$CONDA_DIR/etc/profile.d/conda.sh && conda activate wasb"
59     echo "[gpu] (the LinuxNativeRunner ? backend/pipeline/detectors ? drives extraction once
it lands; M3.5.)"
60

```

17. user (2026-06-20T08:59:50.217Z)

```

1     #!/usr/bin/env bash
2     #
3     # Detect a spare local NVMe and mount it at /scratch for fast video decode.
4     #
5     # DO GPU Droplets ship a SECOND raw NVMe (the "scratch" disk), unmounted by default; a
6     # regular CPU box has only the boot disk ? then we just create /scratch on it. The hot
path
7     # is sync-first (copy video local -> ffmpeg-decode there), so scratch belongs on the
fast
8     # local disk. Point the pipeline at it with  export TMPDIR=/scratch  or  --scratch
/scratch.
9     #
10    # EPHEMERAL by design (wiped on destroy / may not survive a power-cycle) and
deliberately NOT
11    # added to /etc/fstab (a renamed device could block boot). Nothing durable lives here.
12    set -euo pipefail
13    MNT=/scratch
14    mkdir -p "$MNT"
15

```

```

16     root_src=$(findmnt -n -o SOURCE / 2>/dev/null || true)
17     root_disk=$(lsblk -no PKNAME "$root_src" 2>/dev/null | head -1 || true)
18     [ -z "$root_disk" ] && root_disk=$(basename "${root_src:-}" | sed -E 's/p?[0-9]+$/ /')
19
20     cand=""
21     while read -r name type; do
22         [ "$type" = "disk" ] || continue
23         [ "$name" = "$root_disk" ] && continue
24         bytes=$(lsblk -bdno SIZE "/dev/$name" 2>/dev/null | head -1)
25         [ "${bytes:-0}" -lt 10000000000 ] && continue          # skip tiny config drives
26         if lsblk -no MOUNTPOINT "/dev/$name" | grep -q '/'; then continue; fi # skip if
already mounted
27         cand="$name"; break
28     done < <(lsblk -dno NAME,TYPE)
29
30     if [ -n "$cand" ]; then
31         dev="/dev/$cand"
32         echo "[scratch] spare disk: $dev"
33         if ! blkid "$dev" >/dev/null 2>&1; then
34             echo "[scratch] no filesystem -> mkfs.ext4 $dev"
35             mkfs.ext4 -F -q "$dev"
36         fi
37         mountpoint -q "$MNT" || mount "$dev" "$MNT"
38         chmod 1777 "$MNT"
39         echo "[scratch] mounted $dev at $MNT ? $(df -h "$MNT" | awk 'NR==2{print $2}' total,
"$4" free}')")
40     else
41         echo "[scratch] no spare NVMe found ? using $MNT on the boot disk (fine for a CPU
box)"
42         chmod 1777 "$MNT"
43     fi
44
45     echo
46     echo ">> Point the pipeline at scratch: export TMPDIR=$MNT (or pass --scratch $MNT to
backend.worker)"
47

```

18. assistant (2026-06-20T08:59:53.675Z)

Now let me verify a few code-level claims: the FastAPI auth/CORS state, the storage backends, and where the khelsutra site lives. Let me check the main.py CORS config and the worker entrypoint.

19. user (2026-06-20T08:59:55.678Z)

```

backend\main.py:19:from fastapi.middleware.cors import CORSMiddleware
backend\main.py:37:# Enable CORS for the local web interface. allow_origins='' with
allow_credentials=True is
backend\main.py:40:app.add_middleware(
backend\main.py:41:    CORSMiddleware,
backend\main.py:42:    allow_origins=["*"],
backend\main.py:43:    allow_credentials=False,

```

20. user (2026-06-20T08:59:56.253Z)

```

backend\storage\base.py
backend\storage\__init__.py
backend\storage\gcs.py
backend\storage\gdrive.py
backend\storage\local.py
backend\storage\ref.py
backend\storage\s3.py

```

21. user (2026-06-20T08:59:56.253Z)

```
backend\worker\__init__.py
backend\worker\__main__.py
```

22. user (2026-06-20T08:59:59.056Z)

```
30     from backend.utils.paths import resolve_under_root, safe_output_filename,
validate_input_path
31     from backend.config import load_config as load_app_config, AppConfig, StitchingConfig #
new typed config
32     from backend.results import Failure # standardized error/result contract
33     from backend.reporting import emit_indexer_report
34
35     app = FastAPI(title="Sports Highlight Indexer API")
36
37     # Enable CORS for the local web interface. allow_origins='' with allow_credentials=True
is
38     # rejected by browsers per the fetch spec and is an unsafe default to carry into the
39     # auth/tenancy work; this tool uses no cookies, so credentials stay off.
40     app.add_middleware(
41         CORSMiddleware,
42         allow_origins=["*"],
43         allow_credentials=False,
44         allow_methods=["*"],
45         allow_headers=["*"],
46     )
47
48     # Serves static frontend files directly (now under api/ per approved structure)
49     os.makedirs("backend/api/static", exist_ok=True)
50     app.mount("/static", StaticFiles(directory="backend/api/static"), name="static")
51
52     @app.get("/", response_class=HTMLResponse)
53     def serve_index():
54         index_path = "backend/api/static/index.html"
55         if os.path.exists(index_path):
56             with open(index_path, "r", encoding="utf-8") as f:
57                 return f.read()
58         return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/api/static/</h3>"
59
```

23. user (2026-06-20T09:00:00.655Z)

```
backend\main.py:39:# auth/tenancy work; this tool uses no cookies, so credentials stay off.
```

24. assistant (2026-06-20T09:00:04.617Z)

Confirmed: no auth dependency anywhere in main.py (only one comment match for "auth"), CORS is `allow\_origins=["\*"]`. Let me check the khelsutra site location and the KHELSUTRA_PRODUCT_OVERVIEW doc (it's an untracked new file mentioned in git status) for hosting/exposure context.

25. user (2026-06-20T09:00:07.508Z)

```
total 30
drwxr-xr-x 1 avidu 197609  0 Jun 20 14:19 .
drwxr-xr-x 1 avidu 197609  0 Jun 18 11:14 ..
drwxr-xr-x 1 avidu 197609  0 Jun 20 14:29 .git
-rw-r--r-- 1 avidu 197609 271 Jun 20 13:05 .gitattributes
drwxr-xr-x 1 avidu 197609  0 Jun 20 12:43 .github
-rw-r--r-- 1 avidu 197609  60 Jun 10 23:12 .gitignore
-rw-r--r-- 1 avidu 197609 3467 Jun 20 14:19 NEXT_STEPS.md
-rw-r--r-- 1 avidu 197609 925 Jun 20 08:37 README.md
```

```
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:19 docs
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:22 site
drwxr-xr-x 1 avidu 197609    0 Jun 10 23:12 web
---DEPLOY-SCRIPTS-IN-KHELSUTRA---
```

26. user (2026-06-20T09:00:07.574Z)

```
1      # KhelSutra ? Product Overview & Early-User Guide
2
3      *What it does today, how to get the best out of it, and where it's going.*
4      *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6      ---
7
8      ## In one line
9
10     **KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Put a
phone or GoPro
11     on a tripod behind the court, record a session, and KhelSutra automatically finds the
rallies, stitches
12     a **highlight reel**, and gives you a **simple summary** (how many rallies, how much
real play time).
13     No special hardware, no editing, no manual scrubbing.
14
15     ---
16
17     ## The problem we're solving
18
19     Academies and serious players already record a lot of video ? and then almost never
watch it. Hours of
20     footage pile up on phones and memory cards because cutting it into useful clips by hand
is tedious and
21     nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22
23     Most academy halls also run **several courts at once**, so a single recording is a busy,
noisy scene
24     that ordinary video tools choke on. **Pulling the real rallies out of busy, multi-court
footage is the
25     hard problem KhelSutra is built to crack** ? it already works very well on clean,
single-court
26     recordings, and the multi-court case is exactly what we're sharpening now.
27
28     ---
29
30     ## Who it's for
31
32     - **Coaching academies & coaches** ? turn raw session footage into review-ready clips
and reels for
33     teaching, without an editor.
34     - **Serious players** ? get clips of your own rallies to review, keep, and share.
35     - Today's focus is **badminton academies in Bengaluru** ? that's who we're building with
first.
36
37     ---
38
39     ## What it does **today**
40
41     This is what's working right now, in the private web app invited users can try:
42
43     - **Upload a full session or match video** (from a phone on a tripod, or a GoPro). Long
videos are fine.
44     - **Automatic rally detection** ? it finds the rallies and where they start and end, on
its own.
45     - **Highlight reel** ? the rallies stitched into one MP4 you can watch, download, and
share, with a small
```

```

46     "Generated by Khelsutra.guru" mark.
47     - **A simple summary** ? how many rallies were found and how much actual play time there
was.
48     - **Re-use & download** ? process once, come back to it later, download the reel.
49
50     That's the core loop: **record ? upload ? get back a reel + summary.** No editing skill
needed.
51
52     > Like any automatic tool, it can occasionally miss a real rally or include one that
isn't ? telling us
53     > when that happens is genuinely useful feedback, not a sign it's broken (more on that
below).
54
55     ---
56
57     ## How to use it (alpha)
58
59     1. **Your KhelSutra associate helps you get access** to the private web app (it's
invite-only; today the
60         app is in English).
61     2. **Upload your session video.** Large files upload in pieces and can take a while on
Indian uplinks ?
62         that's normal.
63     3. **It processes in the background.** You can close the tab and come back later; you
don't have to sit
64         and watch it.
65     4. **When it's done**, you get the **highlight reel + summary** to watch and download.
66     5. **Re-process anytime.** (Re-running the same video replaces its earlier result rather
than keeping both.)
67
68     ---
69
70     ## How to get the best out of it (capture matters)
71
72     KhelSutra works best on a **steady, full-court recording** ? the same way a coach would
set a camera to
73     review play. A few simple habits make a big difference to the results:
74
75     - **Put the camera behind the court, slightly elevated**, with the **whole court in
frame**.
76     - **Keep it still** ? a fixed tripod beats a handheld or panning shot every time.
77     - **Shoot landscape**, with reasonable lighting.
78     - **Record the whole session** and upload it as-is ? KhelSutra does the cutting.
79
80     Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style
cuts, very dark
81     halls) is the **hardest case** and where results suffer most. When in doubt: fixed
camera, full court,
82     don't touch it. *(Your field contact provides a one-page capture checklist at setup.)*
83
84     ---
85
86     ## The part that makes KhelSutra different: **it gets sharper from real academy
footage**
87
88     This is the heart of the product, and what sets it apart from a generic "AI clipper":
89
90     **The footage that real academies contribute is what makes KhelSutra better over time**
? especially on
91     the busy multi-court halls that are the hard case. When you choose to share your
footage, and when
92     coaches confirm or correct what it found, that growing, labelled library is what we use
to sharpen the
93     tool. It's designed to improve in new versions as that library grows, rather than stay
frozen ? which is

```

94 why early users aren't just customers, they're co-builders.
95
96 ****Your data stays yours ? and sharing is your choice.**** We will ****never**** use your video
to improve the
97 tool without your ****clear, explicit agreement**** ? it is opt-in, never automatic.
****Footage involving**
98 minors is never used to improve the tool and is never published.****** We never publish your
footage; you
99 decide what, if anything, you share.
100
101 ****How your video is handled (plainly):**** these are large videos that need real
processing, so your video
102 is handled securely on our infrastructure to produce your reel. We don't publish it, and
we don't use it
103 to improve the tool without your opt-in.
104
105 > ****Read this honestly:**** today the tool gives you rally detection + reels + summaries.
It does ****not****
106 > retrain itself as you use it, and there is no per-person tracking. The "gets sharper
over time" above is
107 > the model we're building toward, powered by contributed footage ? not a switch that's
already on.
108
109 ---
110
111 **##** Where it's going ? toward an ****AI assistant coach****
112
113 Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
library above ?
114 turns it into a genuine training companion:
115
116 - ****Shot detection**** ? recognising smashes, drops, net shots, serves, and clears within
each rally.
117 - ****Deeper per-rally and per-match analysis**** ? patterns, tendencies, what's working and
what isn't.
118 - ****Per-match history**** ? track progress over weeks and months, not just one clip.
119 - ****Coaching insights**** ? an ****AI assistant coach**** that supports (never replaces) the
real coach,
120 surfacing what to work on next.
121 - ****More sports**** ? table tennis next, then other racket/court sports.
122 - ****Your language**** ? Kannada and Hindi, alongside English.
123
124 > ****Read this honestly:**** the items in this section are the ****direction****, not today's
buttons. What you
125 > can use **today** is rally detection + highlight reels + the play-time summary. The
deeper analysis arrives
126 > as the video library grows ? which is exactly why your footage and feedback matter so
much.
127
128 ---
129
130 **##** What feedback helps us most (for alpha users)
131
132 You don't need to be technical ? just tell us, in your words:
133
134 - ****When it missed a real rally****, or ****grabbed something that wasn't a rally.****
135 - Whether the ****reel** is the right length and order****** for how you'd actually use it.
136 - ****What numbers or details you wish the summary showed.****
137 - Your real ****capture conditions**** ? how many courts were running, camera/phone,
lighting.
138 - Anything that would make this ****genuinely useful for your academy**** (or a reason it
wouldn't be).
139
140 *(Send feedback however is easiest ? your KhelSutra associate will help you with that.)*
141

142 ---
143
144 ## What we're focused on over the next ~4 weeks
145
146 1. **Growing the library of real academy footage** ? especially busy **multi-court**
halls (the hard,
147 valuable case). This is the single biggest lever on quality.
148 2. **Field conversations with academies** ? understanding how coaches and players
actually work with
149 video, and what would make this genuinely useful.
150 3. **Recording sessions and showing academies their own footage** as reels ? **"this is
what it looks
151 like for you."**
152 4. **Sharpening multi-court accuracy** and inviting a few more academies into the
private trial.
153
154 ---
155
156 ## What KhelSutra is **not** (so expectations stay honest)
157
158 - It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough
edges.
159 - It is **not a line-calling, scoring, or umpiring system**, and **not player tracking /
identification**.
160 - **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available
today.**
161 - It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds
insight.
162 - **Pricing isn't set** ? we're still learning what's fair; today's invited use is about
getting it right.
163
164 ---
165
166 ## Quick answers to common questions
167
168 ***"How accurate is it?"***
169 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is
exactly what
170 we're improving. The best answer is to *show* a sample reel rather than quote a number.
171
172 ***"Do I need a special camera or hardware?"***
173 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a
one-page checklist.
174
175 ***"Is my video private? What happens to it?"***
176 We treat your footage as private: we never publish it, and we never use it to improve
the tool without
177 your explicit opt-in. These are large videos that need real processing, so your video is
handled securely
178 on our infrastructure to make your reel. **Footage involving minors is never used to
improve the tool and
179 is never published.**
180
181 ***"Can it tell me the score / who won / call lines?"***
182 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.
183
184 ***"Does it work for doubles and tournaments?"***
185 It finds rallies in doubles too. Tournament footage varies: a clean, fixed, full-court
recording works
186 well; crowded multi-court tournament halls are the hard case we're specifically working
on.
187
188 ***"Will it get better?"***
189 Yes ? that's the whole idea. The more real footage runs through it (with permission) and
the more coaches

190 correct it, the sharper future versions get, and the sooner shot detection and deeper
analysis arrive.
191
192 ***"Which sports?"**
193 Badminton today; table tennis and other racket sports are on the roadmap.
194
195 ***"When can I use it / how do I join?"**
196 It's an invite-only private trial. If you're interested, the best next step is to let us
record a session
197 and show you your own court and players as a reel.
198
199 ---
200
201 *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*
202 *Field associates: the role description is in the Field Associate flyer, and the
conversation script and
203 visit rules are in the separate Field Visit Playbook.*
204

27. assistant (2026-06-20T09:00:10.197Z)

Let me check the khelsutra site directory structure (where /srv/khelsutra-site comes from) and
its deploy/hosting docs to confirm the actual layout.

28. user (2026-06-20T09:00:15.565Z)

C:/Users/avidu/Projects/khelsutra/docs/PER_RALLY_SELECTION.md
C:/Users/avidu/Projects/khelsutra/docs/PORTABILITY.md
C:/Users/avidu/Projects/khelsutra/docs/README.md
C:/Users/avidu/Projects/khelsutra/docs/REQUEST_ACCESS_FORM.md
---SITE---
C:/Users/avidu/Projects/khelsutra/site/make_og.py
C:/Users/avidu/Projects/khelsutra/site/package.json
C:/Users/avidu/Projects/khelsutra/site/public/capture.html
C:/Users/avidu/Projects/khelsutra/site/public/index.html
C:/Users/avidu/Projects/khelsutra/site/public/og.png
C:/Users/avidu/Projects/khelsutra/site/README.md
C:/Users/avidu/Projects/khelsutra/site/schema.sql
C:/Users/avidu/Projects/khelsutra/site/scripts/droplet-smoke.sh
C:/Users/avidu/Projects/khelsutra/site/scripts/parity-check.sh
C:/Users/avidu/Projects/khelsutra/site/server.js
C:/Users/avidu/Projects/khelsutra/site/src/assets.test.js
C:/Users/avidu/Projects/khelsutra/site/src/deploy-config.test.js
C:/Users/avidu/Projects/khelsutra/site/src/index.js
C:/Users/avidu/Projects/khelsutra/site/src/index.test.js
C:/Users/avidu/Projects/khelsutra/site/src/notify.js
C:/Users/avidu/Projects/khelsutra/site/src/server.test.js
C:/Users/avidu/Projects/khelsutra/site/src/store.js
C:/Users/avidu/Projects/khelsutra/site/wrangler.toml

29. user (2026-06-20T09:00:15.703Z)

C:\Users\avidu\Projects\khelsutra\web\README.md:2:Dev: talks to the engine at
http://localhost:8000 via Vite proxy; no auth locally (DEV_USER_EMAIL).
C:\Users\avidu\Projects\khelsutra\NEXT_STEPS.md:11:SQLite** host (`site/server.js`), **live +
browsable on the DO droplet ? http://168.144.126.176:8787**
C:\Users\avidu\Projects\khelsutra\NEXT_STEPS.md:12:(systemd `khelsutra-site`). PRs #1?#8 merged.
Docs: `docs/REQUEST\_ACCESS\_FORM.md`, `docs/PORTABILITY.md`.
C:\Users\avidu\Projects\khelsutra\NEXT_STEPS.md:32:- **Droplet ? production** ? TLS + a
subdomain (`vm.khelsutra.guru` via Caddy) if it should be a real
C:\Users\avidu\Projects\khelsutra\github\workflows\parity.yml:27: VM_URL:
http://168.144.126.176:8787
C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md:113:| P1 | SPA scaffold
(Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to engine `:8000` | P0 | No | ?

```
| ? |
C:\Users\avidu\Projects\khealsutra\docs\PORTABILITY.md:22:| Deploy | git ? Workers Builds | `node
server.js` under systemd |
C:\Users\avidu\Projects\khealsutra\docs\PORTABILITY.md:23:| TLS/DNS | Cloudflare | keep
Cloudflare in front, **or** Caddy/Let's Encrypt |
C:\Users\avidu\Projects\khealsutra\docs\PORTABILITY.md:32:round-trip + cleanup). Verified live on
the owner's droplet (Ubuntu 24.04, isolated `/srv/khealsutra-site`).
C:\Users\avidu\Projects\khealsutra\docs\PORTABILITY.md:40:# put it under systemd (auto-restart) +
a reverse proxy (Caddy/nginx) for TLS,
C:\Users\avidu\Projects\khealsutra\docs\PORTABILITY.md:47: route/origin at the box). Full exit
(own DNS + Caddy TLS) is also supported.
C:\Users\avidu\Projects\khealsutra\site\scripts\parity-check.sh:7:#
CF_URL=https://khealsutra.guru VM_URL=http://168.144.126.176:8787 scripts/parity-check.sh
C:\Users\avidu\Projects\khealsutra\site\scripts\parity-
check.sh:10:VM="${VM_URL:-http://168.144.126.176:8787}"
C:\Users\avidu\Projects\khealsutra\site\scripts\parity-check.sh:20: echo " git -C <repo>
archive HEAD:site | ssh root@<vm> tar -x -C /srv/khealsutra-site/site"
C:\Users\avidu\Projects\khealsutra\site\scripts\droplet-smoke.sh:8:#
[REMOTE_DIR=/srv/khealsutra-site] [PORT=8787] scripts/droplet-smoke.sh
C:\Users\avidu\Projects\khealsutra\site\scripts\droplet-
smoke.sh:12:DIR="${REMOTE_DIR:-/srv/khealsutra-site}"
```

30. assistant (2026-06-20T09:00:19.862Z)

I have all the grounding I need. The khealsutra site is a marketing/request-access site (Node + SQLite, port 8787, systemd `khealsutra-site`, `/srv/khealsutra-site`) ? distinct from the rally engine FastAPI (port 8000, `~/rally`). The droplet `168.144.126.176` hosts the site at port 8787 and the engine lives at `~/rally`. Let me confirm the engine's port and the `web/` dev-proxy auth note to nail the exposure/auth model precisely.

31. user (2026-06-20T09:00:20.333Z)

```
1 SPA scaffold lands here at execution-plan milestone M1 (Vite + React + TS + Tailwind).
2 Dev: talks to the engine at http://localhost:8000 via Vite proxy; no auth locally
(DEV_USER_EMAIL).
3
```

32. user (2026-06-20T09:00:20.774Z)

```
1 # Portability ? Cloudflare now, always-on VM if needed
2
3 > **Status:** design + proven (2026-06-20). **Decision:** keep the site/form host-
portable so
4 > moving off Cloudflare to an owner-run always-on VM ("if Cloudflare acts out") is an
**adapter
5 > swap, not a rewrite.**
6
7 ## The shape that makes it portable
8 - **Logic is runtime-agnostic.** `src/index.js` uses Web-standard `Request`/`Response`
and pure
9 helpers (`validate`, `toRow`, `isBot`, `handleRequest`) ? no host-specific calls.
10 - **All DB coupling is in adapters** (`src/store.js`): `d1Store` (Cloudflare D1) and
`sqliteStore`
11 (node:sqlite / better-sqlite3). `handleRequest(request, store)` takes the store by
injection.
12 - **One schema, one SQL.** `schema.sql` and the `INSERT` are plain SQLite, so D1 and a
VM's SQLite
13 file are byte-for-byte compatible ? zero dialect change. (Postgres later = minor
placeholder tweaks.)
14
15 ## Two hosts, same code
16 | | Cloudflare (now) | Always-on VM (fallback) |
17 |---|---|---|
18 | Entry | `src/index.js` (Worker `fetch`) | `server.js` (node:http) |
```

```

19 | Static | `env.ASSETS` | `server.js` serves `public/` |
20 | Storage | `d1Store(env.DB)` ? D1 | `sqliteStore(db)` ? `node:sqlite` file |
21 | Country | `cf-ipcountry` header | optional (GeoIP / null) |
22 | Deploy | git ? Workers Builds | `node server.js` under systemd |
23 | TLS/DNS | Cloudflare | keep Cloudflare in front, **or** Caddy/Let's Encrypt |
24
25 **Enforced, not asserted.** `node --test src/` runs BOTH paths and CI
(`.github/workflows/test.yml`,
26 Node 24) runs them on every PR, so the repo stays compatible with both hosts:
27 - `src/index.test.js` ? the Cloudflare Worker + D1 logic.
28 - `src/server.test.js` ? **boots the real `server.js`** (node:http + node:sqlite) and
round-trips it
29 over HTTP into SQLite (the same proof we run on the droplet, automated).
30
31 For a live check against a real VM: `SSH_TARGET=root@host scripts/droplet-smoke.sh` (scp
+ boot +
32 round-trip + cleanup). Verified live on the owner's droplet (Ubuntu 24.04, isolated
`/srv/khelsutra-site`).
33
34 ## VM deploy runbook ("if Cloudflare acts out")
35 Node 22+ only ? **no `npm install`** (node:sqlite is built in).
36 ```
37 # on the VM:
38 git clone ? && cd khelsutra/site # or scp the site/ folder
39 PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
40 # put it under systemd (auto-restart) + a reverse proxy (Caddy/nginx) for TLS,
41 # then point DNS at the VM. ~1 hour; logic + schema unchanged.
42 ```
43
44 ## Honest notes
45 - "Cloudflare acts out" usually means **Workers/D1 pricing or limits, not DNS** ? so the
cheapest
46 move is to keep Cloudflare DNS/CDN in front and shift only the **compute** to the VM
(point a
47 route/origin at the box). Full exit (own DNS + Caddy TLS) is also supported.
48 - You can run **both** (Cloudflare primary + VM warm standby) since the schema is
identical; a
49 periodic `access_requests` export keeps them in sync.
50 - `server.js` is intentionally dependency-free; if you later want Postgres or a
framework (Hono runs
51 on Workers *and* Node), it slots in behind the same `store` + handler.
52

```

33. assistant (2026-06-20T09:00:23.663Z)

I now have complete, verified grounding across all the requested docs, deploy scripts, and code. Let me compile the structured output.

34. user (2026-06-20T09:01:38.003Z)

Structured output provided successfully

35. assistant (2026-06-20T09:01:52.846Z)

I've mapped the hosting, exposure, locality, desktop vision, and droplet layout and returned the findings via StructuredOutput.

Key verified facts:

- ****Exposure/auth:**** Single-tenant, private/invite-only. The engine FastAPI ships NO app-layer auth and open CORS (`allow\_origins=["\*"]`, backend/main.py:42-43). The only intended gate is Cloudflare Access (email-OTP allowlist) at the edge + a Cloudflare Tunnel that keeps the box IP unexposed ? both in HOSTING_PLAN.md (status PROPOSAL, not yet ratified/built).
- ****Locality:**** VIDEO_LOCALITY_MODEL.md's ladder L3?L2?L1?L0?L4; "timestamps are the product, video is dead weight"; recommended alpha default = L2 proxy upload (review UI streams the 1080p

proxy). Today's L3 flow keeps originals forever (retention is an acknowledged gap).

- ****Droplet:**** ONE shared CPU-only DO droplet 168.144.126.176 runs BOTH the engine (`~/rally`, FastAPI :8000) and the khelsutra marketing/request-access site (`/srv/khelsutra-site`, Node+SQLite :8787, systemd `khelsutra-site`). The GPU box is a separate, not-yet-built cattle droplet (RTX 4000 Ada, non-blr1 region, NVMe scratch, native-WASB conda env). Secrets live only on the box (chmod-600 key files).

- ****Desktop:**** Yes ? DESKTOP_APP_PLAN.md envisions a locally-installed cross-platform app (CLI/installer MVP ? Tauri GUI, Python/FastAPI bundled as a localhost sidecar via PyInstaller, NativeWasbRunner with no WSL). It IS the L0 / LocalDesktop ComputeBackend; everything is gated on the P0 CPU/MPS feasibility spike.

- ****Guardrails:**** no app auth yet, edge-only intended gate, unsolved retention, minors excluded from training, paid-LLM no-train, per-video cost ceiling, secrets-on-box-only.

One note: `git pull --ff-only` failed ("Cannot fast-forward to multiple branches") and the current branch `feat/native-wasb-runner` is diverged, so per your repo-freshness convention I proceeded read-only on the local checkout ? findings may lag the remote.